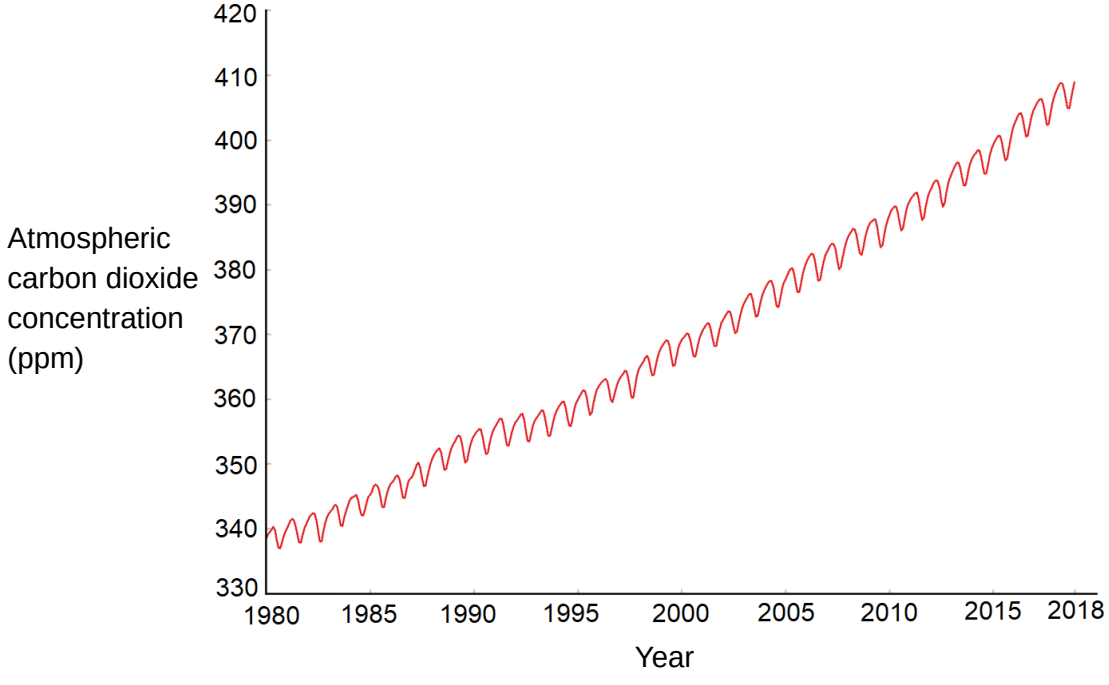
Structured questions

|!|BCELB0360000001|!| ★ **STSE**

The combustion of fossil fuels releases large amounts of carbon dioxide into the atmosphere. The graph shows trends in atmospheric carbon dioxide concentrations from 1980 to 2018.



(a) With reference to the graph, the average carbon dioxide concentration in the atmosphere fluctuates during a year in which the concentration in summer is lower while higher in winter. Suggest an explanation to this phenomenon.

(3 marks)

(b) Explain how the increased carbon dioxide concentration in the atmosphere results in global warming. (2 marks)

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(c) Suggest why global warming would lead to rises in sea levels in many coastal cities. (1 mark)

(d) Apart from the combustion of fossil fuels, state and explain ***one*** human activity that leads to the increase in atmospheric carbon dioxide concentration. (3 marks)

##

(a) As the day length in summer is longer / air temperature in summer is higher, (1)
the overall photosynthetic activity of plants is higher. (1)
Plants take in more carbon dioxide from the atmosphere, (1)
thus lowering the atmospheric carbon dioxide concentration.

(b) Carbon dioxide in the atmosphere traps / absorbs heat radiation reflected from the ground, (1)
leading to an increase in the atmospheric temperature. (1)

(c) Global warming raises sea water temperatures, causing the ice sheets in the polar regions to melt / causes thermal expansion of sea water. (1)

(d) Deforestation (1)
leads to a decrease in the overall photosynthetic activity and thus a reduction in the removal of carbon dioxide from the atmosphere. (1)
As a result, the atmospheric carbon dioxide concentration increases, leading to an increased greenhouse effect. (1)

(9 marks)

##

	Skill assessed	Mark(s)
(a)	Applying concepts	3
(b)	Understanding basic concepts	2
(c)	Understanding basic concepts	1
(d)	Understanding basic concepts	3

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|!|BCELB0360000002|!| ★

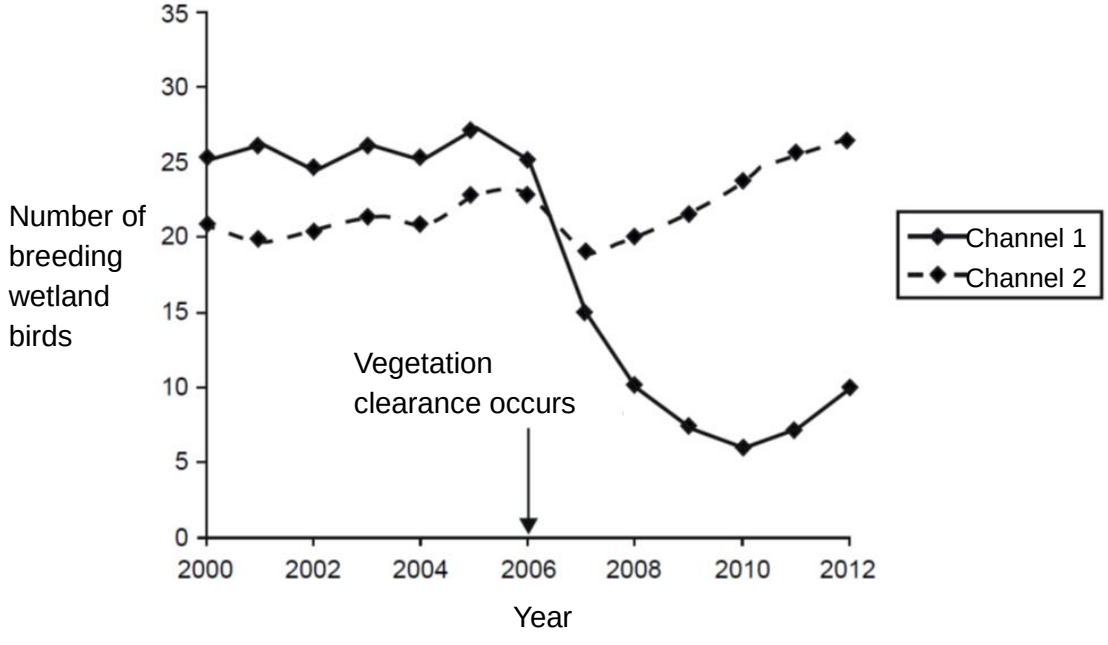
CCEA GCE AS Biology Assessment Unit AS 2 Jun 2014 Q8

Drainage channels are important in preventing excessive waterlogging and flooding of low-lying farmland. Left undisturbed, they become overgrown with plants and function less effectively. However, such overgrown drainage channels form important wildlife habitats.

The sides of two overgrown drainage channels were cleared of vegetation using two different strategies:

- Channel 1 had both sides cleared.
- Channel 2 had only one side cleared.

The total number of breeding wetland birds at each channel was surveyed for a number of years before and after clearance, which occurred in 2006.



(a) Describe the trends in bird numbers for both channels from 2000 to 2012. (3 marks)

(b) Suggest possible reasons for the changes in the number of birds in each channel after 2006. (2 marks)

(c) In order to promote biodiversity on farms, agri-environment schemes often suggest reducing the amount of artificial fertilizer used on farmland.

(i) Describe ***two*** ways in which excessive use of artificial fertilizer might have a negative effect on biodiversity on the land. (2 marks)

(ii) Using the information at the beginning of this question, suggest how the use of artificial fertilizers on low-lying farmland might actually increase biodiversity around drainage channels. (2 marks)

##

(a) Any ***three*** of the following: (3)

- Before 2006 / vegetation clearance, bird numbers remained relatively constant/similar at both channels. (1)
- After 2006, bird numbers decreased for both channels/channel 1 showed a more marked decline. (1)
- After 2006, bird numbers at channel 1 continued to decline but began to rise from 2010 onwards. (1)
- After 2007, bird numbers at channel 2 increased beyond their initial value. (1)

(b) For channel 1, the clearance of both sides reduces food availability/shelter/nest sites. (1)

For channel 2, a greater range of habitats/food/shelter is created. (1)

(c) (i) Increased soil erosion/decrease of soil crumb structure leads to death of soil organisms/plants (1)

promotes the growth of a few plant species at the expense of many others. (1)

(ii) Water plant growth will be promoted/channels will become overgrown (1)

provides more food/shelter for wetland birds. (1)

(9 marks)

##

	Skill assessed	Mark(s)
(a)	Analysing data	3
(b)	Analysing data	2
(c)(i)	Understanding basic concepts	1
(c)(ii)	Understanding basic concepts	3

|!|BCELB0360000003|!| ★ **STSE** **Cross topic**

CCEA GCE AL Biology Assessment Unit A2 1 Jun 2013 Q7

Lough Neagh is the most highly eutrophic lake in Ireland, enriched with high levels of nitrate and phosphate.

(a) The following table shows the sources of phosphate entering Lough Neagh in the year 2000.

Source	Phosphate entering Lough Neagh		Additional notes
	Tonnes	%	
Towns	129	25.4	value decreasing
Industry	6.8	1.3	value relatively static
Septic tanks	62	12.2	consequence of large number of rural farms with septic tanks and inefficient soakaway systems
Agriculture	310.7	61.1	proportion increasing as other sources decrease or remain static

(i) Suggest how phosphate (and nitrate) pollution from septic tanks can be decreased. (1 mark)

_____

_____

(ii) Much of the agricultural contribution to phosphate (and nitrate) pollution comes from the inappropriate use or overuse of artificial fertilizer.
Describe how the use of artificial fertilizer can lead to pollution of waterways and a subsequent reduction in aquatic life. (3 marks)

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(iii) Describe ***two*** distinct ways in which farmers can reduce the level of water pollution caused by artificial fertilizer. (2 marks)

(b) In Lough Neagh one of the species that was found in high numbers is *Anabaena*, a blue-green alga that is capable of fixing nitrogen.
Using this information, suggest why phosphate, rather than nitrate, is thought to have been mainly responsible for the problems in Lough Neagh. (2 marks)

##

(a) (i) Treat the sewage in a sewage treatment works before discharge / the use of reed beds / explanation of more efficient soakaway system (e.g. create terraces by ploughing across land / herringbone drainage) (1)
(Or other acceptable answers)

(ii) Nitrate / phosphate leaches into waterways leading to algal blooms. (1)
Decomposition of dead algae / increased bacterial population / aerobic respiration by bacteria occurs. (1)
Aquatic life dies due to shortage of oxygen. (1)

(iii) Any ***two*** of the following: (2)

- Calculation of application levels to match crop needs / so supply does not exceed demand. (1)
- Only apply fertilizer immediately prior / or during periods of vigorous growth. (1)
- Do not spray when rain is forecasted. (1)
- Do not spray near waterways. (1)
- Use more organic fertilizer / use less artificial fertilizer. (1)

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(Or other acceptable answers)

(b) *Anabaena* produces its own ammonia (nitrate) / amino acids. (1)

Nitrate is less likely to be limiting (than phosphate) / does not require nitrate from the environment. (1)

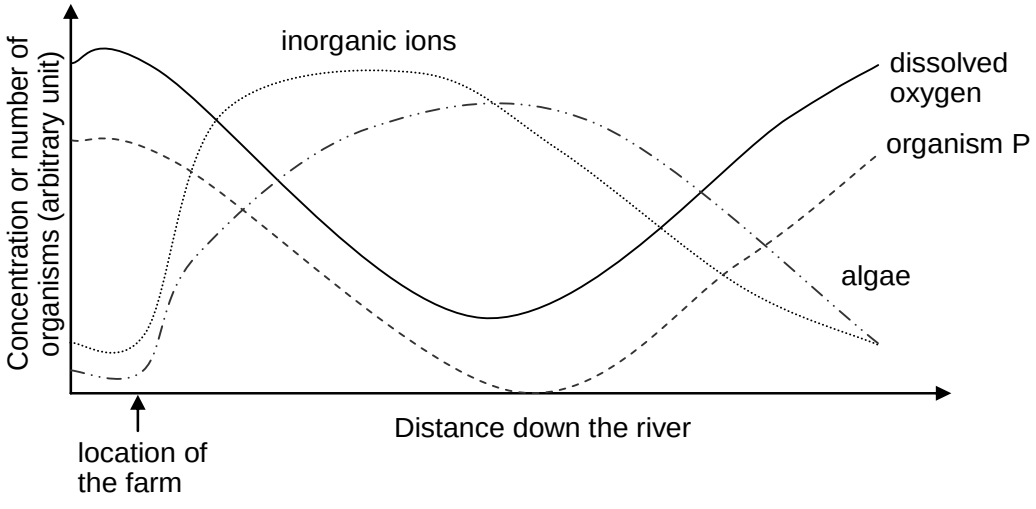
(8 marks)

##

	Skill assessed	Mark(s)
(a)(i)	Applying concepts	1
(a)(ii)	Understanding basic concepts	3
(a)(iii)	Applying concepts	2
(b)	Applying concepts	2

|!|BCELB0360000004|!| ★

The graph below shows the concentrations of inorganic ions, concentrations of dissolved oxygen and the numbers of algae and organism P along a river.



(a) What is the possible cause of the change in concentration of inorganic ions downstream of the farm? Give ***one*** example of the inorganic ions. (3 marks)

(b) With reference to the graph, explain why the dissolved oxygen concentration decreases significantly with an increase in the level of inorganic ions. (3 marks)

(c) Suggest ***two*** reasons for the decrease in the number of organism P downstream of the farm. (2 marks)

(d) Explain why the dissolved oxygen concentration resumes to normal after some distance down the river. (1 mark)

##

(a) The farmers apply chemical fertilizers on the farmland. (1)
The fertilizers are then leached into the river by rain / irrigation water. (1)
Example of inorganic ion: Nitrates (1)
(Or other acceptable answers)

(b) Algae grow rapidly with the high concentration of inorganic ions/under eutrophication. (1)
At night, algae carry out respiration only. (1)
The large population of algae consumes a large amount of dissolved oxygen in the water. (1)

(c) A number of organism P die of suffocation because oxygen in the water may be used up by the large algal population. (1)
They may also be killed by the toxins produced by some algal species. (1)

(d) Fast-flowing (river) water will gain oxygen through mixing/turbulence. (1)
(Or other acceptable answers)

(9 marks)

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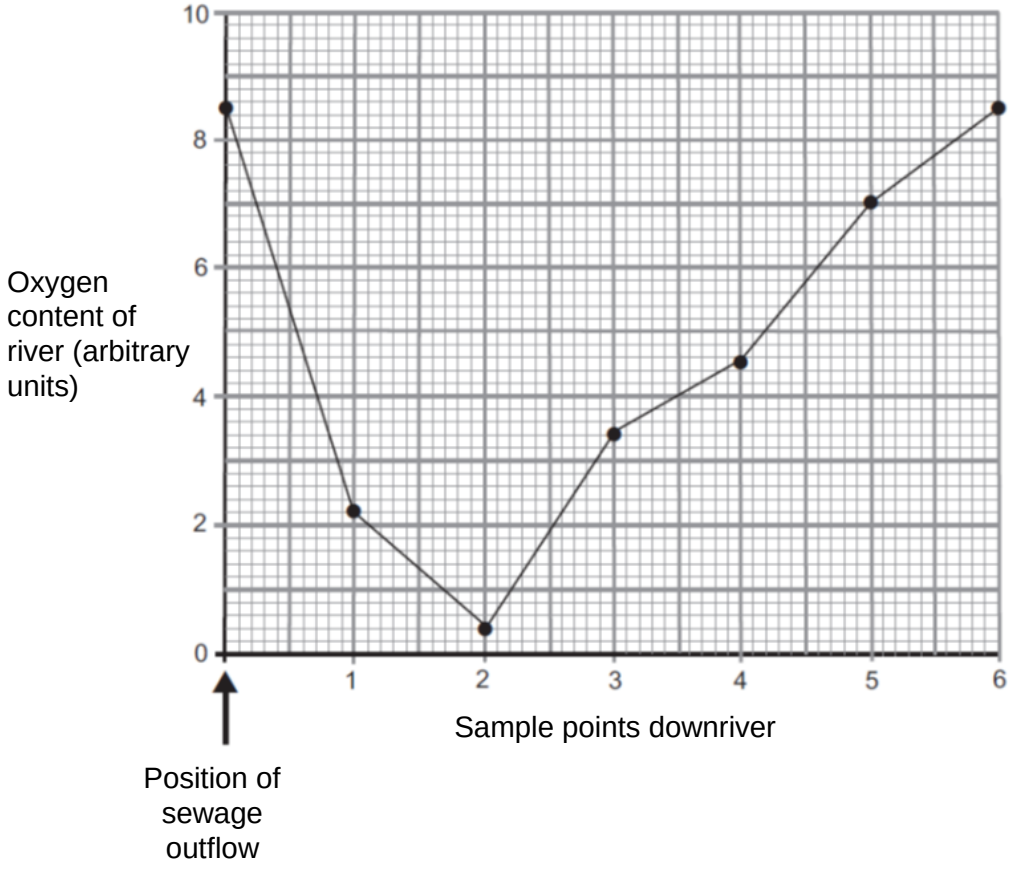
##

	Skill assessed	Mark(s)
(a)	Applying concepts	2
	Understanding basic concepts	1
(b)	Analysing data	1
	Understanding basic concepts	2
(c)	Applying concepts	2
(d)	Applying concepts	1

|!|BCELA0360000005|!| ★★ **SI**

CCEA GCE AL Biology Assessment Unit A2 1 Jun 2011 Q3

A river, otherwise unpolluted, was sampled from the source of a sewage outflow. Oxygen content of the river was measured at the point of outflow and at intervals downriver. The results are shown in the graph below.



(a) (i) Calculate the percentage decrease in the oxygen content between the sewage outflow and sample point 2. (Show your working.) (2 marks)

(ii) Explain this decrease in oxygen content. (3 marks)

(b) Suggest ***two*** reasons for the increase in oxygen content from sample points 2 to 6. (2 marks)

(c) Identify the sample point from the graph where you would expect to find the lowest biodiversity in the river. Explain your answer. (2 marks)

(d) Aquatic invertebrates can act as 'indicator species' in monitoring river pollution. Suggest why these species are a better indicator of pollution than chemical tests for oxygen levels. (2 marks)

##

(a) (i) 8.5 – 0.4 (1)

8.1 ÷ 8.5 × 100 = 95.3% (1)

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(ii) Any ***three*** of the following: (1)

- Organic content of sewage is fed on by saprophytic microbes/decomposers/bacteria feed on sewage (1)
- Population explosion of these organisms (1)
- Increased respiration / aerobic activity of these organisms (causes fall in oxygen levels) / use of oxygen in respiration (1)
- Increased turbidity of water would reduce photosynthesis (1)

(b) Any ***two*** of the following: (2)

- Bacterial / saprophytic population declines/less sewage available (1)
- Reeds remove sewage / produce oxygen (1)
- Dilution downstream (1)
- Increased photosynthesis (produces more oxygen due to less turbidity/nitrate-rich water) (1)
- Fast flowing (river) water will gain oxygen through mixing / turbulence (1)

(Or other acceptable answers)

(c) Sample point 2 (1)

Most river species are unable to survive in very low oxygen levels. / Anaerobic conditions will only favour the growth of a small number of species. (1)

(d) Any ***two*** of the following: (2)

- Each group of indicator species can tolerate particular conditions / particular level of anoxic conditions / pollution. (1)
- Indicator species act as long-term monitors of oxygen levels. (1)
- Chemical tests only at specific times / give measurement at specific times. (1)
- Measuring oxygen level using a chemical test does not indicate its effect on aquatic life. (1)

(11 marks)

##

	Skill assessed	Mark(s)
(a)(i)	Mathematical skills	2
(a)(ii)	Applying concepts	1
(b)	Applying concepts	2
(c)	Analysing data	1
	Applying concepts	1
(d)	Critical thinking	2

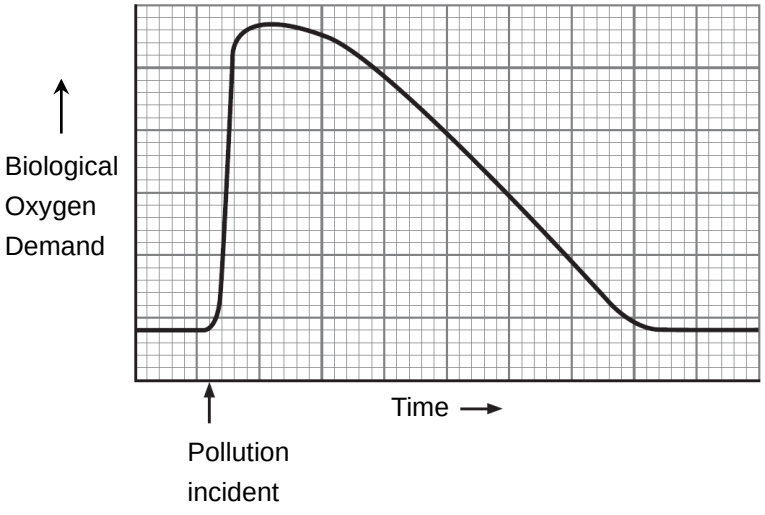
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|!|BCELA0360000006|!| ★★ **SI**

CCEA GCE AL Biology Assessment Unit A2 1 Jun 2015 Q5(a)–(b)

In order to promote environmental conservation, human activity needs to be managed and monitored. There are many aspects to this, including reducing pollution levels and conserving existing habitats.

(a) The graph below shows how an incident of slurry (淤漿) pollution in a small lake affected the Biological Oxygen Demand (BOD) (生物需氧量) over a period of time.



(i) Explain the immediate effect of the slurry pollution on BOD in the lake. (3 marks)

(ii) Artificial fertilizer entering waterways also affects BOD, although the sequence of events and the timescale are different. Give ***one*** similarity and ***two*** differences between the effects of artificial fertilizer and slurry on BOD in waterways. (3 marks)

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(b) Indicator species can be used to monitor pollution levels in waterways. A number of different species of aquatic invertebrates (e.g. bloodworms and stoneflies) can be used for this purpose.

Suggest ***two*** advantages of using invertebrates, which occupy lower trophic levels in the aquatic food web, as indicator species, rather than top predator fish such as pike. (2 marks)

##

(a) (i) Slurry is rich in organic/decomposable material, (1)
causing population explosion of bacteria. (1)
The aerobic respiration by bacteria increases BOD. (1)

(ii) Similarity: Both increase BOD. (1)
Differences: Timescales with artificial fertilizer/eutrophication are longer. (1)
With eutrophication, source of decomposable matter is algae / with slurry, source is animal waste. (1)

(b) Any ***two*** of the following: (2)

- Greater number, hence the quantitative data more meaningful (1)
- Easier to trap / monitor / count (1)
- Bioaccumulation affects invertebrates more / invertebrates more sensitive / invertebrates have a greater range of sensitivity (1)

(Or other acceptable answers)

(8 marks)

##

	Skill assessed	Mark(s)
(a)(i)	Analysing data	3
(a)(ii)	Applying concepts	3
(b)	Applying concepts	2

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|!|BCELB0360000007|!| ★ **STSE**

The table shows the effect of different concentrations of inorganic nitrogen fertilizer on the yield of lettuce.

Inorganic nitrogen fertilizer applied (kg / hectare)	**Yield (tonnes / hectare)**
50	9
100	14
150	18
200	23
250	27
300	27

(a) Suggest ***one*** mineral ion that may be present in the inorganic nitrogen fertilizer. (1 mark)

(b) Describe and explain the effect of adding inorganic nitrogen fertilizer on the yield of lettuce. (4 marks)

(c) (i) The excessive use of inorganic fertilizers can lead to a reduction in the biodiversity of farmland. Explain why it is so. (2 marks)

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(ii) Explain why a reduction in biodiversity will cause problems for agriculture in the future. (2 marks)

##

(a) Nitrate (1)
(Or other acceptable answers)

(b) As the amount of inorganic fertilizer added increased from 50 to 250 kg/hectare, the yield of lettuce increased sharply (1)
because nitrogen is needed for the synthesis of proteins/nucleic acids in plants. (1)
As the amount of inorganic fertilizer added increased from 250 to 300 kg/hectare, the yield remained constant / did not increase (1)
because other minerals / environmental factors become the limiting factor. (1)

(c) (i) Inorganic fertilizers promote the growth of one / few crop species (e.g. crop species). The other plant species are out-competed as a result of competition from crop species. (1)
The disappearance of these plant species may disrupt the food chains. (1)
(Or other acceptable answers, e.g. the reduction in soil quality / humus over time so plants cannot grow)

(ii) Environment / agricultural requirements may change in the future. (1)
A reduction in biodiversity means the loss of genetic diversity / variation in wild population. Some of the lost alleles / genes may have been useful to cope with these changes in the environment. (1)
(Or other acceptable answers, e.g. fewer pollinators may lead to reduction in crop yield, loss of pest predators may cause pest problems, etc.)

(9 marks)

##

	Skill assessed	Mark(s)
(a)	Understanding basic concepts	1
(b)	Analysing data	2
	Applying concepts	2
(c)(i)	Applying concepts	2
(c)(ii)	Applying concepts	2

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|!|BCELB0360000008|!| ★ **Cross topic**

DDT is a pesticide. It is insoluble in water but soluble in lipids. In the 1950s, large amounts of DDT were sprayed to prevent the spread of malaria in some countries. The table below shows the results of an analysis of the concentration of DDT in the body tissues of some bird species during that time period.

Bird species	Main contents of diet	DDT concentration in body tissues (ppm)
A	Insects and plant material	0.6
B	Fish	14.9
C	Plant material	0.3
D	Fish	6.9
E	Small mammals	4.5
F	Plant material	0.2

(a) Suggest why spraying DDT was effective in preventing malaria in humans.
(2 marks)

(b) With reference to the table, describe and explain the relationship between the diet content and the concentration of DDT in the body tissues of different bird species.
(4 marks)

(c) A small population of mosquito is found to possess a gene that makes them resistant to DDT. Suggest how this gene could make the mosquitoes resistant to DDT. (2marks)

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##

(a) DDT can kill mosquitoes, which are the vector of malaria. (1)

With fewer mosquitos, the spread of malaria can be prevented. (1)

(b) Animal-eating (carnivorous) birds (e.g. B, D and E) have higher DDT concentration in body tissues than plant-eating (herbivorous) birds (e.g. C and F). (1)

Animal-eating birds occupy a higher trophic level than plant-eating birds. (1)

DDT cannot be metabolized or excreted once taken in, and thus can accumulate along the food chains. (1)

Therefore, animals at a higher trophic level accumulated more DDT when they feed on organisms at lower trophic levels. (1)

(c) The gene produces certain proteins / enzymes, (1)

which prevent the absorption of DDT / break down DDT molecules. (1)

(Or other acceptable answers)

(8 marks)

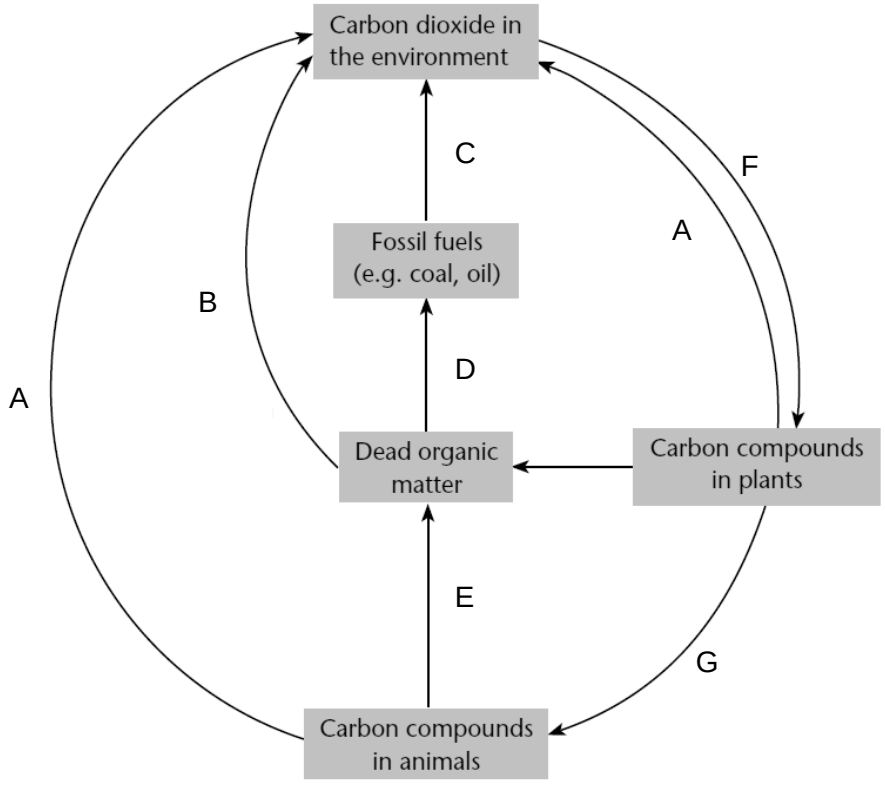
##

	Skill assessed	Mark(s)
(a)	Understanding basic concepts	2
(b)	Analysing data	1
	Applying concepts	3
(c)	Applying concepts	2

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|!|BCELB0360000009|!| ★ **STSE** **Cross topic**

The diagram below shows the carbon cycle.



(a) Name the processes that shown by the arrows A and F. (2 marks)

(b) Using the letters in the diagram, state ***one*** process in the carbon cycle that involves the actions of microorganisms. (1 mark)

(c) Over the last few decades, the carbon dioxide concentration in the atmosphere has been rising, resulting in global warming. With reference to the carbon cycle, suggest how this has happened. (5 marks)

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(d) Global warming has led to an increase in the concentration of salt (salinity) in soil water in some parts of the world.

(i) Suggest how global warming could lead to an increase in soil salinity.

(2 marks)

(ii) Explain why most plants cannot survive when the soil salinity is high.

(2 marks)

##

(a) A: Respiration (1)
F: Photosynthesis (1)

(b) B (1)

(c) Due to human activities, process C occurs at a higher rate, producing a larger amount of carbon dioxide. (1)
Also, deforestation reduces the number of trees, which leads to a reduction in the rate of process F. This slows down the removal of carbon dioxide from the atmosphere. (1)
As the addition of carbon dioxide into the atmosphere is faster than the removal, the carbon dioxide concentration in the atmosphere increases. (1)
With more carbon dioxide, the atmosphere traps more heat radiated from the Earth's surface. (1)
This causes an increase in atmospheric temperature. (1)

(d) (i) The increase in atmospheric temperature (1)
causes more water to evaporate from the soil / more frequent flooding / a reduction in rainfall. (1)

(ii) The water potential of the soil water becomes lower than that of the roots. (1)
Water moves out of the roots by osmosis. (1)

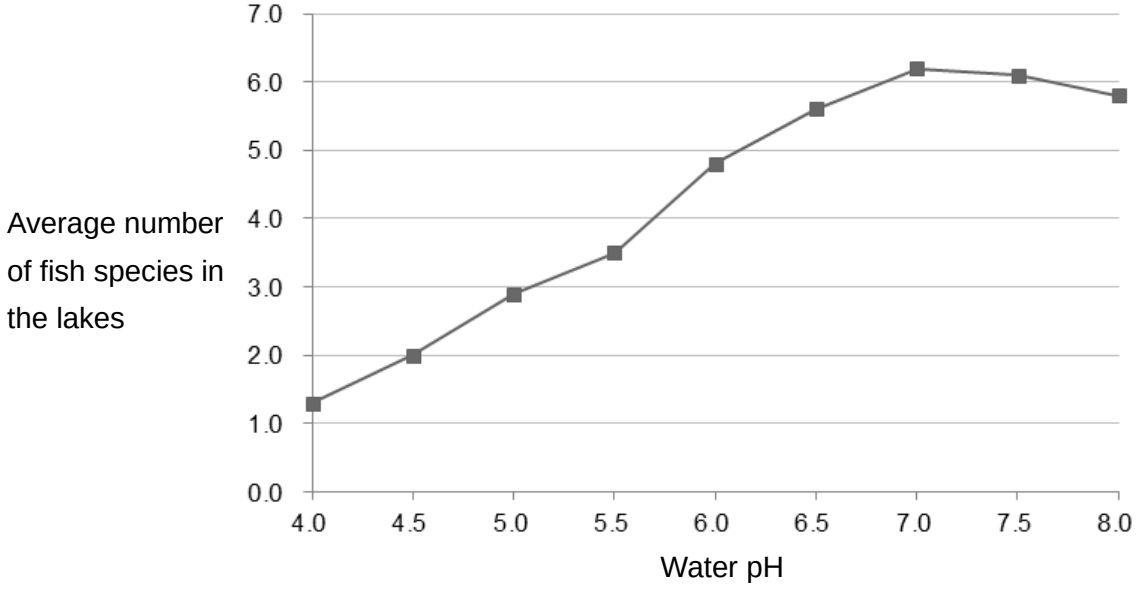
(12 marks)

##

	Skill assessed	Mark(s)
(a)	Analysing data	2
(b)	Analysing data	1
(c)	Applying concepts	5
(d)(i)	Applying concepts	2
(d)(ii)	Applying concepts	2

|!|BCELB0360000010|!| ★ **STSE**

Acid rain is one form of pollution that damages many lakes, ponds and rivers. The graph below shows the effect of water pH on the average number of fish species found in lakes in country X.



(a) (i) State ***one*** air pollutant that causes acid rain. (1 mark)

_____

(ii) Suggest ***one*** source of the air pollutant stated in (i). (1 mark)

_____

(iii) Suggest how the air pollutant can cause acid rain. (2 marks)

_____

_____

_____

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(b) With reference to the graph,

(i) describe the effects of the changes in water pH on the average number of fish species in the lakes. (3 marks)

(ii) explain the effects of acid rain on the fish species in the lakes. (2 marks)

##

(a) (i) Sulphur dioxide / nitrogen oxides (1)

(ii) Burning of fossil fuels (1)

(iii) Sulphur dioxide / nitrogen oxides dissolve(s) in rainwater to form sulphuric acid / nitric acid, which is a strong acid. (1)

The acid makes rainwater much more acidic than normal, with a pH value lower than 5.6. (1)

(b) (i) Increase in pH above 7.0 decreases the number of fish species (1)

Small decline in number of fish species between pH 6.0 and 7.0 (1)

Large decline in the number of fish species between pH 4.0 and 6.0 (1)

(ii) Acid rain reduces the pH of the lake water. (1)

The invertebrates / smaller fish that are sensitive to acidity may not survive. (1)

Less food is available to the fish in the lake. (1)

(Or other acceptable answers) (Max. 2)

##

	Skill assessed	Mark(s)
(a)(i)	Understanding basic concepts	1
(a)(ii)	Understanding basic concepts	1
(a)(iii)	Understanding basic concepts	2
(b)(i)	Analysing data	3
(b)(ii)	Applying concepts	2

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|!|BCELC0360000011|!| **STSE**

Read the passage below and answer the questions that follow:

> The average temperature of the Earth has risen 0.8 °C in the past century. It is expected that the global temperature will rise another 1.1 °C to 2.9 °C over the next hundred years. The major causes of global warming are the over-consumption of fossil fuel and extensive deforestation. These activities emit large amount of greenhouse gases which result in global warming.

(a) Name ***one*** greenhouse gas. (1 mark)

_____

(b) Explain why the activities mentioned in the passage contribute to global warming. (5 marks)

_____

_____

_____

_____

_____

(c) State ***two*** consequences of global warming. (2 marks)

_____

_____

_____

(d) Suggest why fossil fuel is considered as non-renewable energy. (1 mark)

_____

(e) To relieve the global warming condition, some countries have developed alternative energy sources to reduce the use of fossil fuel. Suggest ***one*** example of alternative energy source. (1 mark)

_____

##

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(a) Carbon dioxide / methane / ozone (Or other acceptable answers) (1)

(b) The over-consumption of fossil fuel produces a large amount of carbon dioxide / greenhouse gases. (1)

Fewer trees are available to absorb carbon dioxide from the atmosphere after deforestation. (1)

Less carbon dioxide is removed from the atmosphere. (1)

The carbon dioxide concentration in the atmosphere increases. (1)

More heat is trapped by the greenhouse gases / increases the greenhouse effect. (1)

(c) Any ***two*** of the following: (1 mark each) (2)

- increased melting of icebergs / flooding
- loss of habitats
- spread of infectious diseases due to warmer climate

(Or other acceptable answers)

(d) The rate of regeneration of the fossil fuels is much lower than that of consumption. (1)

(e) Wind power / nuclear power / geo-heat / hydroelectric power (Or other acceptable answers) (1)

(10 marks)

##

	Skill assessed	Mark(s)
(a)	Understanding basic concepts	1
(b)	Applying concepts	5
(c)	Understanding basic concepts	2
(d)	Understanding basic concepts	1
(e)	Understanding basic concepts	1

|!|BCELC0360000012|!| **STSE**

Monoculture is a common practice nowadays. A single crop species is grown over a large area of field for years.

(a) Explain why the soil would become infertile after years of monoculture.(2 marks)

(b) Explain why monoculture also leads to an increase in pest population. (2 marks)

(c) Because of the increase in the pest population, the farmer applies high doses of pesticides to control the pests. A bird caught near a field is examined. It was found that the body of the bird contained a high concentration of a type of pesticide residue which is extensively used by the farmers in the field nearby.

(i) Suggest ***two*** properties of pesticide residue which make the pesticide accumulate in the body of the bird. (2 marks)

(ii) Why is there a high concentration of pesticide residue in the body of the bird? (2 marks)

##

(a) Some crops may require specific nutrients and minerals (1)
After years of monoculture, these nutrients become depleted and the soil becomes infertile. (1)

(b) When the same type of crop is grown continuously, the pests that feed on the crop would have constant and sufficient food supply. (1)
The pests can thus grow and reproduce quickly. (1)

(c) (i) Any ***two*** of the following: (1 mark each) (2)
- high fat solubility
- stable chemical
- cannot be broken down
- cannot be excreted

(ii) The pests that feed on the crops obtain the pesticide residues from the crops. (1)
The birds that feed on those pests will have the pesticides accumulated in the body of the birds. (1)

(8 marks)

##

	Skill assessed	Marks
(a)	Understanding basic concepts	2
(b)	Understanding basic concepts	2
(c)(i)	Understanding basic concepts	2
(c)(ii)	Applying concepts	2

|!|BCELC0360000013|!| **Cross topic**

The following table shows the average life span of people living in four poor and developing countries in 1985 and 2005.

Country	Average life span of people in 1985 (years)	Average life span of people in 2005 (years)
A	43	48
B	58	40
C	54	50
D	48	48

(a) Which country has an increase in the average life span of its people from 1985 to 2005? (1 mark)

(b) Suggest ***two*** reasons to explain why the average life span of human is increasing worldwide. (2 marks)

(c) AIDS is a disease that shortens the life span of a human. Which country has the most serious problem of AIDS since 1985? Explain your answer. (2 marks)

(d) In poor countries, the risk of diarrhoea is high. Suggest ***two*** ways to reduce the risk of having diarrhoea. (2 marks)

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##

(a) Country A (1)

(b) Any ***two*** of the following: (1 mark each) (2)

- The improvement of agricultural technology increases the food supply.
- The improvement in the treatments of diseases increases our life expectancy.
- The improved sanitation / awareness of public health increases our life expectancy.

(Or other acceptable answers)

(c) Country B, (1)

because AIDS can lead to a larger number of death at a younger age. (1)

(d) Any ***two*** of the following: (1 mark each) (2)

- to keep the drinking water clean by sanitation
- to educate the people in such countries about personal hygiene
- to cook food thoroughly before consumption

(Or other acceptable answers)

(7 marks)

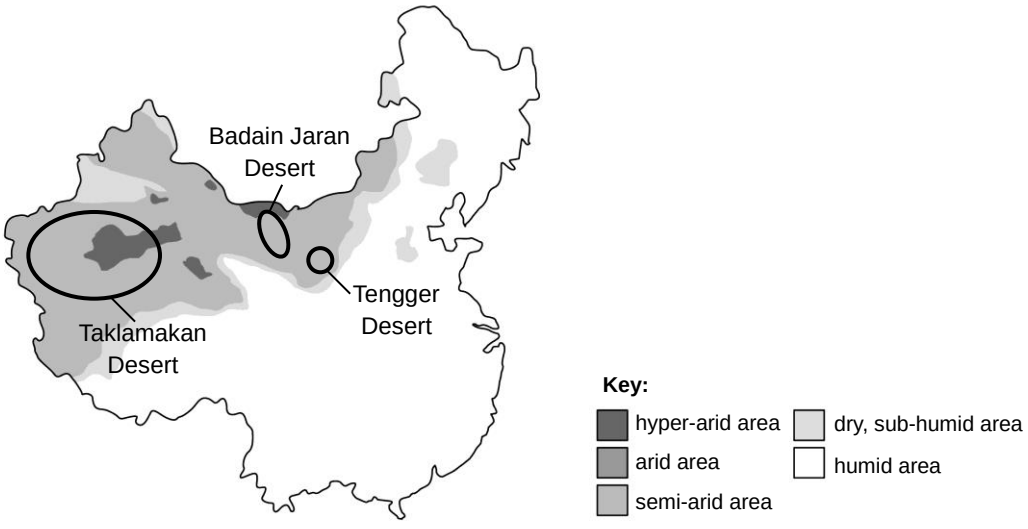
##

	Skill assessed	Mark(s)
(a)	Analysing data	1
(b)	Understanding basic concepts	2
(c)	Analysing data	2
(d)	Understanding basic concepts	2

|!|BCELB0360000014|!| ★ **STSE**

Deserts are found mainly in the northwestern part of China. Most of the land is farmed and the annual rainfall is low in this area. The local farmers allow their livestock to roam and graze free over the semi-arid grassland. Deserts in China have expanded in the past decade at an annual rate of about 1500 square miles. The Chinese government is working for saving the once-fertile farmland from becoming deserts. The figure below shows the desertification-prone areas in China.

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(a) What climatic factor leads to desertification? (1 mark)

(b) State ***two*** causes that lead to the desertification of a semi-arid grassland.(2 marks)

(c) State ***two*** harmful effects caused by desertification. (2 marks)

(d) Suggest ***two*** ways to prevent desertification in dry areas. (2 marks)

(e) Soya beans and peanuts are leguminous plants. Explain how growing these crops would help increase the fertility of the soil and thus increase the yields of other crops such as maize. (3 marks)

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##

(a) Decreasing rainfall (1)

(b) Any ***two*** of the following: (1 mark each) (2)

- Vegetation cover is lost due to overgrazing.
- Soil erosion occurs.
- The evaporation of water from soil increases.
- Soil becomes infertile and cannot support plant.

(Or other acceptable answers)

(c) Any ***two*** of the following: (1 mark each) (2)

- Biodiversity decreases.
- Sandstorm occurs.
- Land is unsuitable for cultivation, may lead to a decrease in food production.

(Or other acceptable answers)

(d) Any ***two*** of the following: (1 mark each) (2)

- Grow plants that are drought-resistant
- Set up proper irrigation system
- Control the grazing of livestock
- Add stones / gravels over the ground to retain water

(Or other acceptable answers)

(e) These crops have nitrogen-fixing bacteria in their root nodules. (1)

Nitrogen-fixing bacteria carry out nitrogen fixation to convert nitrogen in soil air to ammonium compounds. (1)

The increase in the content of ammonium compounds in soil improves the soil fertility and allows other crops to grow properly. (1)

(10 marks)

##

	Skill assessed	Mark(s)
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	2
(c)	Understanding basic concepts	2
(d)	Understanding basic concepts	2
(e)	Understanding basic concepts	3

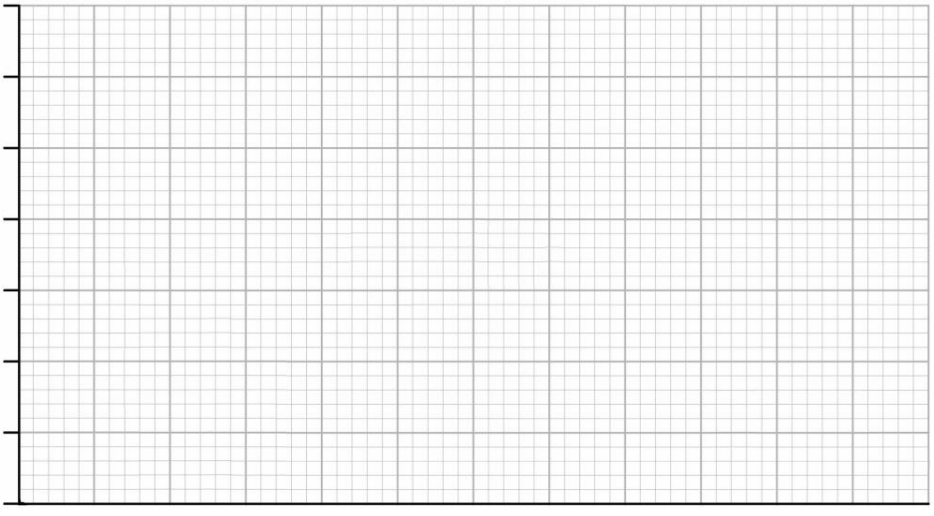
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|!|BCELB0360000015|!| ★

In order to increase the wheat yield, farmers usually add nitrate fertilizers to their field. The table below shows the wheat yields when different amounts of nitrate fertilizer were added to the field.

Amount of fertilizers added (kg per hectare)	0	50	100	150	200	250
Wheat yield (tonnes per hectare)	3	11	20	31	30	31

(a) (i) Using the graph paper below, plot a graph to show the relationship between the amount of fertilizer added and the wheat yield in the field. (3 marks)



(ii) Using the above plotted graph, state the least amount of nitrogen fertilizers that needs to be added per hectare in order to obtain the maximum yield of wheat. (1 mark)

_____

(b) What is the function of nitrate in the growth of wheat plant? (1 mark)

_____

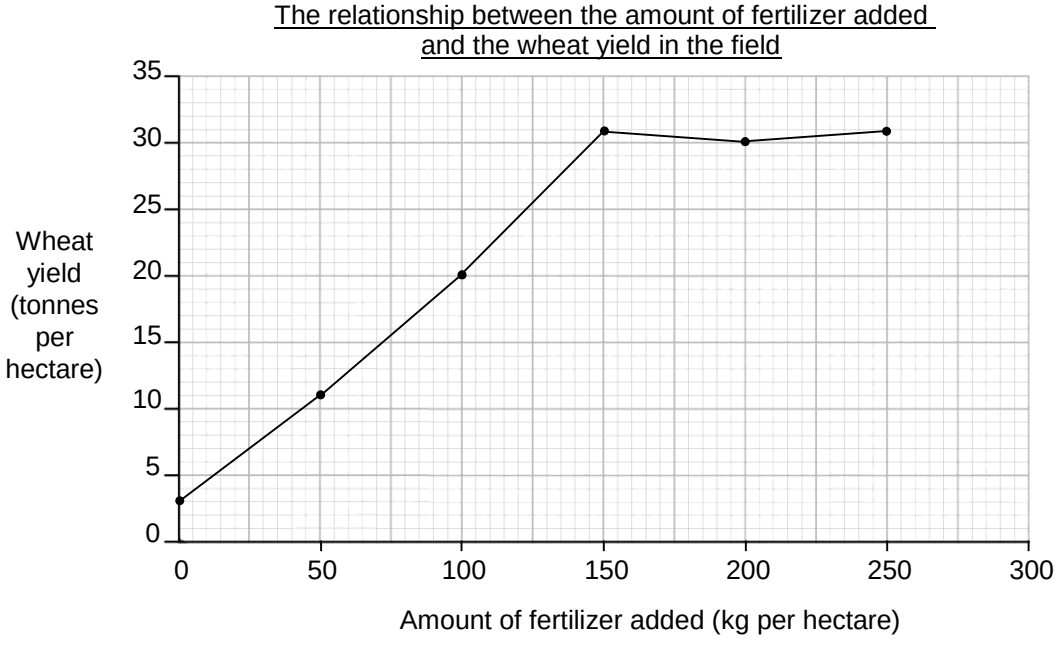
_____

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(c) Algal bloom may be arisen by using excess nitrate fertilizers because the fertilizers may leach from the soil into rivers and lakes. Describe and explain the effects of the nitrate fertilizers on the aquatic ecosystem. (5 marks)

##

(a) (i)



Correct axes (1)
Correct plotting and joining of all points (1)
Appropriate title (1)

(ii) 150 kg / hectare (1)

(b) for protein / amino acid synthesis (1)

(c) More algae grow at the water surface due to an increased supply of nitrate. (1)
Light is blocked by the algal bloom and plants beneath cannot carry out photosynthesis. (1)
When the algae die, the decomposers break down the dead bodies and use up oxygen. (1)

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Less oxygen is available in water and aquatic organisms die because they suffocate. (1)

The food chain is disturbed when the producers increase / decrease in numbers. (1)

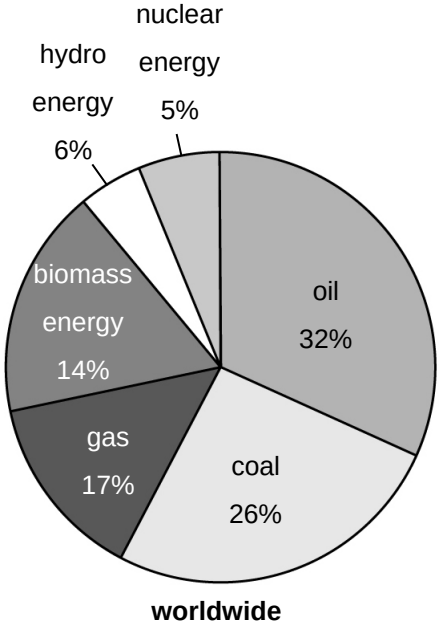
(10 marks)

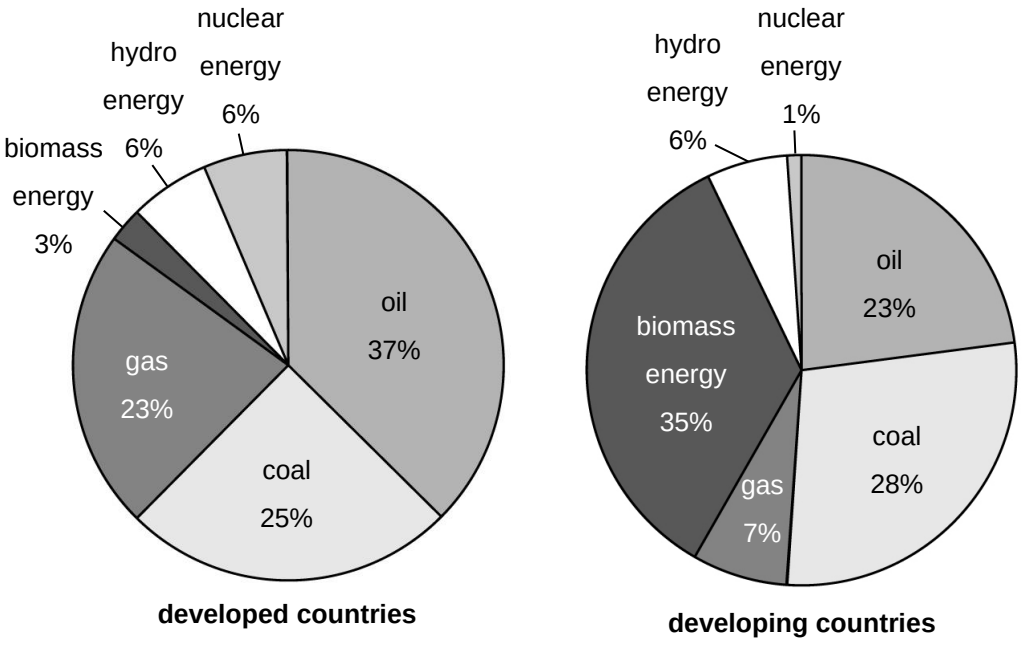
##

	Skill assessed	Mark(s)
(a)(i)	Presenting information	3
(a)(ii)	Analysing data	1
(b)	Understanding basic concepts	1
(c)	Understanding basic concepts	5

|!|BCELC0360000016|!|

The pie charts below show the distribution of various energy sources worldwide, in the developed countries and in the developing countries.





(a) Which energy sources are renewable? Explain why these energy sources are renewable. (3 marks)

(b) Calculate the percentage of worldwide energy sources that are obtained from fossil fuels. Show your working. (2 marks)

(c) According to the pie charts, how many times is the use of biomass in developing countries of that in developed countries? Suggest ***two*** reasons for the difference. (3 marks)

##

(a) Biomass energy (1)
Hydro energy (1)

This is because these energy sources are produced by nature continuously and can be replaced after use. (1)

(b) Percentage of gas + Percentage of coal + Percentage of oil
= 17% + 26% + 32% (1)
= 75% (1)

(c) About 12 times (1)
Any ***two*** of the following reasons: (1 mark each) (2)
- Wood is more abundant in developing countries.
- Biomass fuel is more available in developing countries.
- The cost of biomass fuel is cheaper.
(Or other acceptable answers)

(8 marks)

##

	Skill assessed	Mark(s)
(a)	Analysing data	2
	Understanding basic concepts	1
(b)	Mathematical skills	2
(c)	Mathematical skills	1
	Understanding basic concepts	2

|!|BCELC0360000017|!|

(a) The number of livestock has increased significantly in both developed and developing countries that led to the degradation of grasslands by grazing of livestock.

(i) Describe how an increased number of livestock can bring about the degradation of grasslands. (2 marks)

(ii) Suggest ***two*** methods to reduce the degradation of grasslands by grazing. (2 marks)

(b) Wood is a major resource for both developing and developed countries. Nowadays, more wood is cut down and collected in the rural areas and brought to town for sale.

(i) Suggest ***two*** uses of wood in developing and developed countries. (2 marks)

(ii) State ***three*** problems that will arise from the increase in the cutting down of woods from rural areas. (3 marks)

##

(a) (i) Overgrazing by the livestock damages the grasses / removes the vegetation. (1)

This speeds up soil erosion in the lands. (1)

(ii) Any ***two*** of the following: (1 mark each) (2)

- Do not allow the livestock to roam free over the grassland
- Keep the livestock in small fields to allow time for the recovery of grassland
- Apply fertilizers to increase the growth of grass / vegetation

(Or other acceptable answers)

(b) (i) As fuel (1)

As timber for the production of paper or furniture (1)

(Or other acceptable answers)

(ii) Any ***three*** of the following: (1 mark each) (3)

- Increased soil erosion
- Global warming caused by greenhouse effect
- Shortage of fuel
- Loss of habitats for wildlife

(Or other acceptable answers)

(9 marks)

##

	Skill assessed	Marks
(a)(i)	Understanding basic concepts	2
(a)(ii)	Understanding basic concepts	2
(b)(i)	Understanding basic concepts	2
(b)(ii)	Understanding basic concepts	3

|!|BCELB0360000018|!| ★

Lake Baikal is a large lake in Russia. It is surrounded by mountains and conifer forests. The clearing of the forests for the collection of timber results in soil erosion and landslides. Because of the large-scale logging, large amounts of wood and plant debris have been brought into the lake and become rotten. During this period of time, aquatic organisms in the lake have decreased sharply in number and diversity.

(a) Suggest ***three*** reasons to explain why forests can decrease soil erosion and landslides. (3 marks)

(b) Explain why the aquatic organisms in the lake decrease sharply in number and diversity during the clearing of the forests. (5 marks)

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(c) How may large-scale logging lead to an increase in the average global temperature? (4 marks)

##

(a) Any ***three*** of the following: (1 mark each) (3)

- Tree roots can bind soil particles together and stabilize the soil.
- Tree canopy reduces the impact of rain on soil.
- Trees slow down the flow of surface runoff.
- Trees reduce the impact of wind on soil.
- Fallen leaves / debris on the soil surface can protect the soil.

(Or other acceptable answers)

(b) Plant debris enriches nutrient levels in the lakes as it rots. (1)
This leads to algal bloom. (1)
Large numbers of algae die and are broken down by bacteria / decomposers. (1)
The breakdown of algae consumes oxygen in the lake water. (1)
Sharp decrease in oxygen level causes other organisms to die of suffocation. (1)

(c) Trees can maintain the balance of oxygen and carbon dioxide in the atmosphere. (1)
With fewer trees, less carbon dioxide is absorbed from the atmosphere. (1)
Carbon dioxide is a greenhouse gas and can lead to greenhouse effect. (1)
As a result, deforestation intensifies the greenhouse effect and the world will become warmer. (1)

(12 marks)

##

	Skill assessed	Marks
(a)	Understanding basic concepts	3
(b)	Applying concepts	5
(c)	Understanding basic concepts	4

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|!|BCELB0360000019|!| ★

(a) When a country becomes more developed, its demand for agricultural products will increase. Give ***one*** reason for this. (1 mark)

(b) ***Figure 1*** shows some consequences of intensified agriculture.

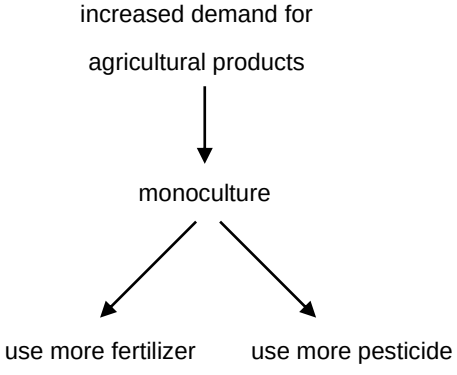


Figure 1

(i) Explain the term monoculture. Give ***one*** advantage for farmers to practice monoculture. (2 marks)

(ii) Explain why monoculture results in the extensive use of fertilizers and state ***one*** ecological problem that can be caused by the extensive use of fertilizers. (2 marks)

(iii) Explain why monoculture results in the extensive use of pesticides and state ***one*** ecological problem that can be caused by the extensive use of pesticides. (2 marks)

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(c) Some people in tropical rainforests practise shifting cultivation. They cut and burn the trees and vegetations in an area in order to clear the area for farming. Ash collected is then used to enrich the soil and crops are grown on the cleared land for a few years. When the soil fertility of the land decreases, farmers will leave the land for many years to regenerate and clear another area for farming.

Figure 2 shows the effects of shifting cultivation on soil fertility at low and high population densities of the crops.

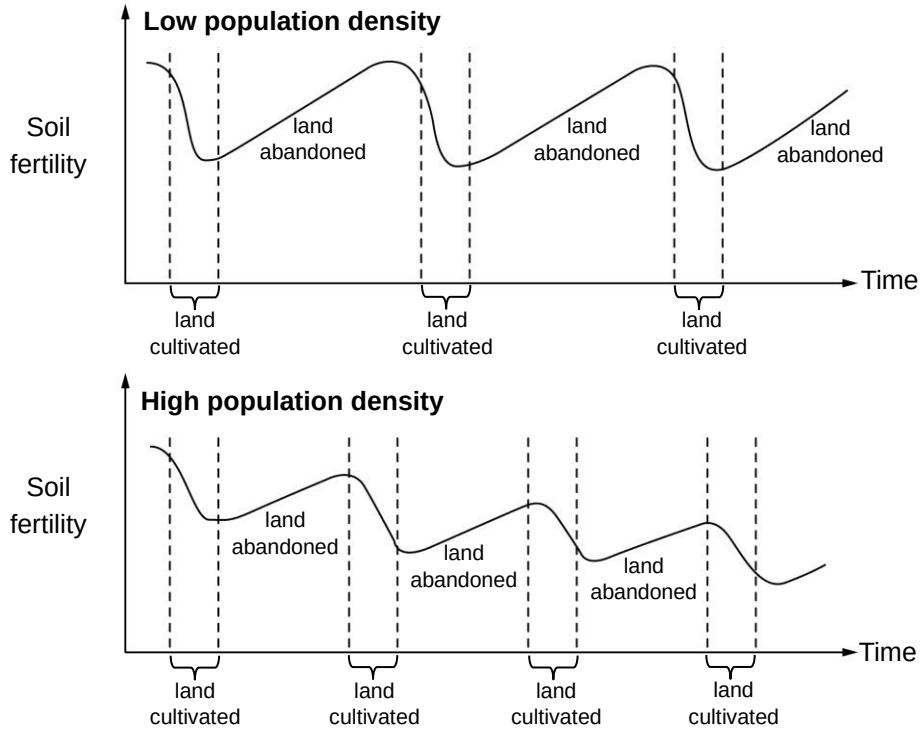


Figure 2

(i) With reference to ***Figure 2***, state the effect of higher population density of crops on soil fertility. (1 mark)

_____

(ii) Suggest reasons for the change in the soil fertility stated in (c)(i). (2 marks)

_____

_____

##

(a) A more developed country has an increasing population that increases the demand of agricultural food. (1)

(b) (i) Monoculture is the agricultural practice of growing a single crop species over a large area for a number of years (1)

Advantage of monoculture: Farmers can keep growing the most profitable crops over a large area for years. (1)

(ii) Monoculture may result in the depletion of certain minerals in the soil. If fertilizer is not added, the soil may become infertile. (1)

Any ***one*** of the following ecological problems: (1)

- Fertilizers may be washed away by rain into rivers or lakes and lead to chemical pollution.
- Extensive use of fertilizers can lead to soil erosion.

(Or other acceptable answers)

(iii) Monoculture involves one single crop that pests will spread faster once they are attacked. (1)

Therefore, farmers rely more heavily on pesticides.

Any one of the following ecological problems: (1)

- It disturbs the ecological balance.
- It promotes the development of new strains of pests which are resistant to the pesticide by natural selection.

(Or other acceptable answers)

(c) (i) Soil fertility declines when there is a higher population density of crops. (1)

(ii) The time for soil fertility to recover is reduced. (1)

Regeneration of forest decreases and so less ash is formed when it is cleared again. (1)

(10 marks)

##

	Skill assessed	Mark(s)
(a)	Understanding basic concepts	1
(b)(i)	Understanding basic concepts	2
(b)(ii)	Understanding basic concepts	2
(b)(iii)	Understanding basic concepts	2
(c)(i)	Analysing data	1
(c)(ii)	Applying concepts	2

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|!|BCELC0360000020|!| **STSE** **Cross topic**

There are 60 to 80 million tonnes of fish caught by commercial fishing boats every year. Overfishing exerts a severe effect on the stocks of many fish species that are caught for food.

(a) State ***two*** signs that indicate the occurrence of overfishing. (2 marks)

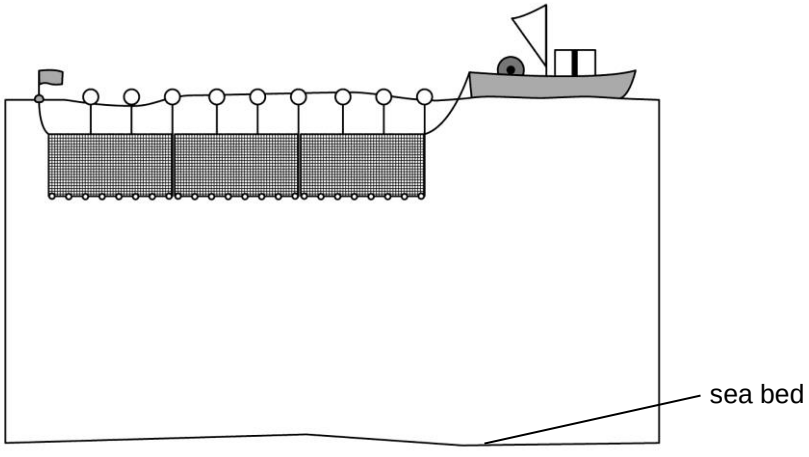
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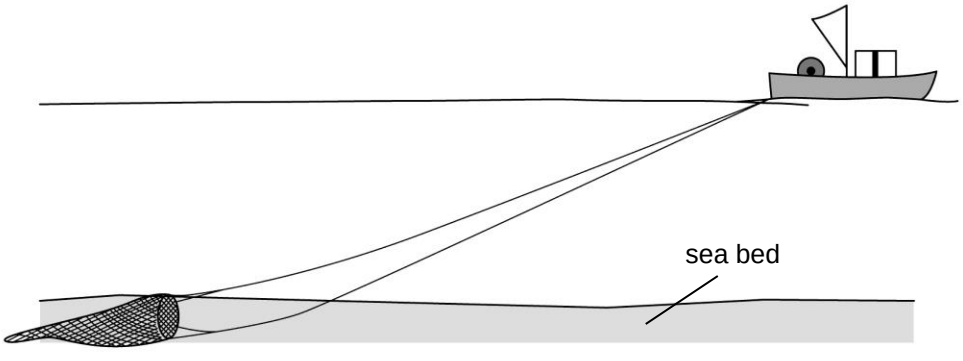
(b) Although fish is a renewable resource, it is vulnerable to depletion. Therefore, fishing activities have to be controlled and managed properly to ensure sustainable fishery.

Drift net fishing and bottom trawling are two main methods used by fishing fleets. The diagrams below show the two methods.



Drift net fishing

(Drift nets are hanged below the water surface and traps fish by the gills.)



Bottom trawling

(Bag-shaped net is dragged over the sea-bed to catch bottom-living fish)

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(i) State how controlling the mesh size of drift nets could help regulate fish stocks and control overfishing. (3 marks)

(ii) Suggest ***two*** ways by which bottom trawls can damage the marine ecosystem. (2 marks)

(c) Suggest ***two*** measures other than controlling the mesh size to help the sustainable fisheries. (2 marks)

##

(a) Smaller numbers of fish in one catch under the same effort (1)
Fewer large / mature fish in one catch (1)

(b) (i) Use drift nets with larger mesh size (1)
so that only large fish are caught / immature fish can escape. (1)
This allows the small / immature fish to become sexually mature and reproduce. (1)

(ii) Any ***two*** of the following: (1 mark each) (2)
- The seabed is damaged.
- Predators of fish may die due to shortage of preys which are caught by trawls.
- Some consumers are caught by trawls and this disturbs the ecological balance.

(Or other acceptable answers)

(c) Any ***two*** of the following: (1 mark each) (2)
- Promoting aquaculture
- Enforcing the fishing moratorium

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- Banning destructive fishing practices such as dredging and dynamite fishing
- Setting up artificial reefs to restore habitats

(Or other acceptable answers)

(9 marks)

##

	Skill assessed	Marks
(a)	Understanding basic concepts	2
(b)(i)	Understanding basic concepts	3
(b)(ii)	Understanding basic concepts	2
(c)	Understanding basic concepts	2

|!|BCELB0360000021|!| ★

Figure 1 shows the average costs of producing electricity from different energy sources in 2003.

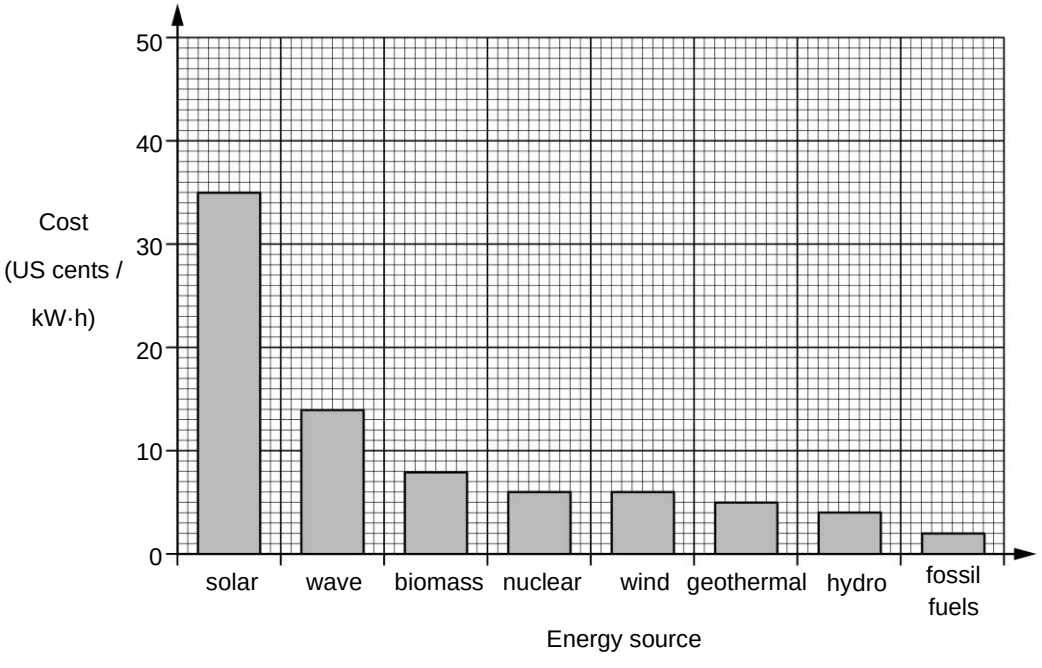


Figure 1

(a) Which of the above is a non-renewable energy source? (1 mark)

(b) Which energy source is the most expensive to be used to produce electricity? Suggest a reason to explain why. (2 marks)

(c) State ***two*** limitations of producing electricity with wave. (2 marks)

(d) State ***one*** advantage and ***one*** disadvantage of using nuclear energy sources to generate electricity. (2 marks)

(e) ***Figure 2*** shows the amounts of oil discovered and demanded in the world.

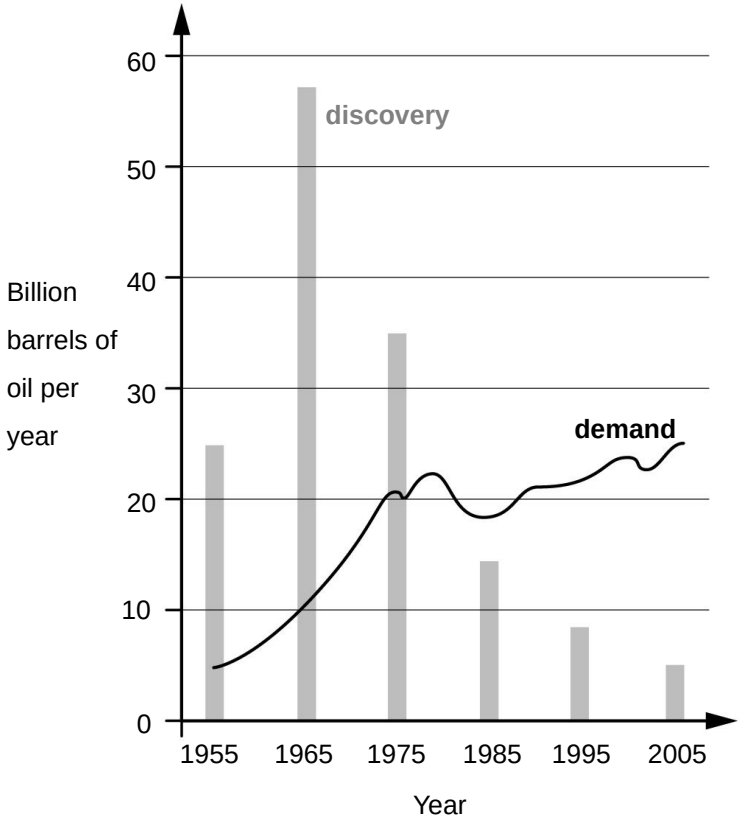


Figure 2

(i) Describe the changes of the demand for oil between 1955 and 2005. (3 marks)

(ii) According to ***Figure 2***, describe how the energy crisis has arisen. (2 marks)

##

(a) Fossil fuels / nuclear power (1)

(b) Solar energy (1)
The manufacturing of solar panels is expensive. (1)
(Or other acceptable answers).

(c) Any ***two*** of the following: (1 mark each) (2)
- The set-up of wave power plant may disturb the marine environment.
- Only seashores can be used to build up wave power plants.
- Wave power plant may affect the lives of the local residents.
(Or other acceptable answers)

(d) Any ***one*** of the following advantage: (1)
- It causes much less air pollution than power stations using fossil fuels because much less carbon dioxide / greenhouse gas is emitted.
- It is a relatively cheaper source of energy.
- It uses only small amounts of raw materials (e.g. uranium) to generate electricity.

(Or other acceptable answers)
Any ***one*** of the following disadvantage: (1)
- Radioactivity from nuclear energy source may cause cancers in people.
- It is dangerous if used irresponsibly by nations / terrorists.
- Nuclear waste may leak into nearby seas / water courses and disturb the ecosystem.

(Or other acceptable answers)

(e) (i) There is a steep rise from around 5 billion barrels in 1955 to 20 billion barrels in 1975. (1)
There is a slight drop from 1980 to 1985. (1)
However, there is persistent increase to 25 billion barrels by 2005. (1)

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(ii) The discovery of oil is decreasing in recent years, (1)
while the demand for oil is increasing and higher than the discovery. (1)

(12 marks)

##

	Skill assessed	Mark(s)
(a)	Analysing data	1
(b)	Understanding basic concepts	2
(c)	Understanding basic concepts	2
(d)	Understanding basic concepts	2
(e)(i)	Analysing data	3
(e)(ii)	Analysing data	2

|!|BCELB0360000022|!| ★ **Cross topic**

(a) ***Figure I*** shows the world map showing the distribution of main fishing areas, which are indicated by the grey areas.

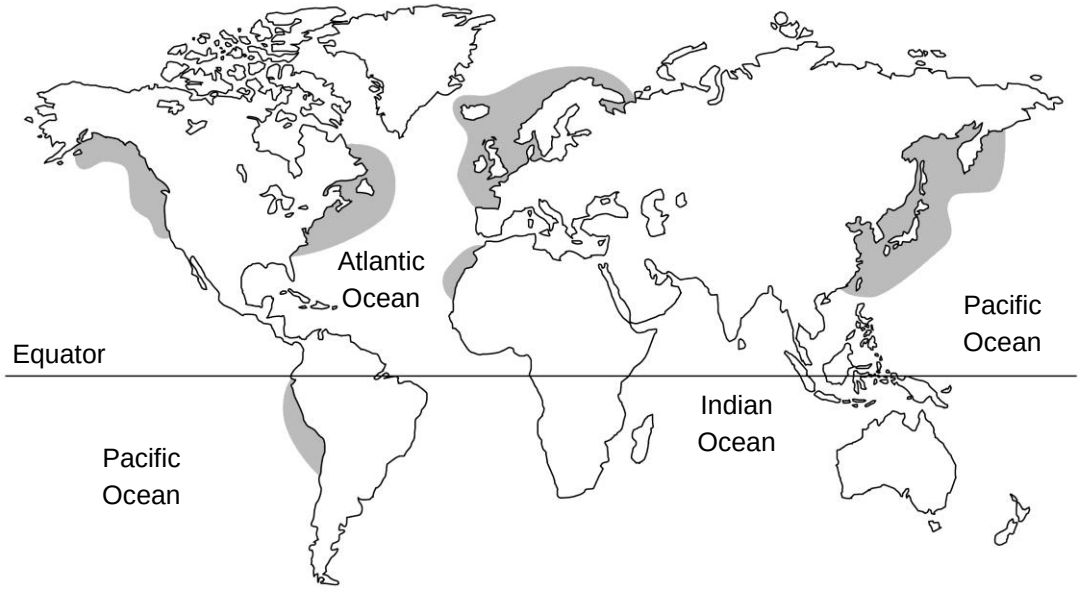


Figure I

According to ***Figure I***, state ***three*** features of the locations of the main fishing areas. (3 marks)

(b) The human causes of overfishing include:

1. fishing boats with a large size
2. advanced technology for locating shoals of fish
3. bigger nets in use

State how each of the above factors leads to overfishing. (3 marks)

(c) ***Figure II*** shows the total weight of fish caught in the world from 1950 to 2000.

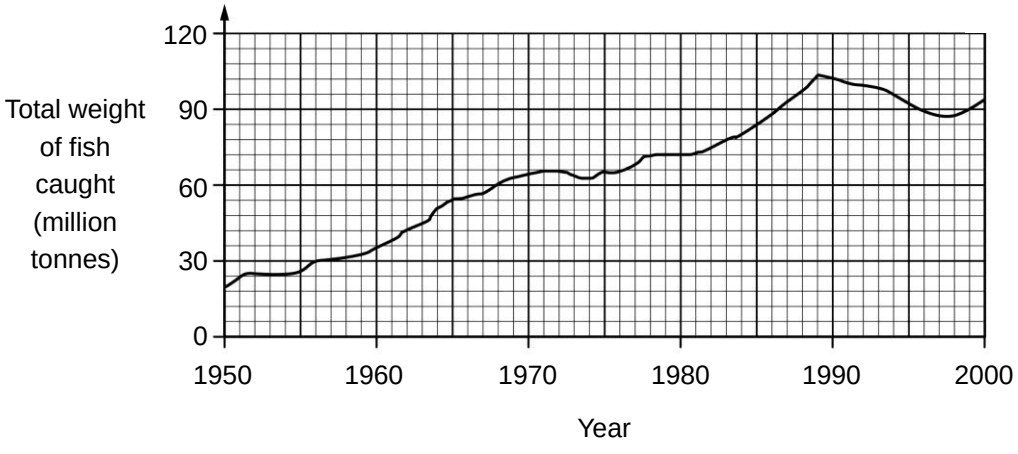


Figure II

(i) State and explain the evidence from ***Figure II*** which shows the occurrence of overfishing. (2 marks)

(ii) Suggest ***three*** ways to conserve and increase the marine fish stocks.(3 marks)

(iii) Suggest ***two*** reasons to explain why attempts to conserve marine fish stocks are not always successful. (2 marks)

_____

_____

_____

##

(a) Any ***three*** of the following: (1 mark each) (3)

- Most are located in the northern hemisphere.
- Most are located in temperate regions.
- They are located both on the western or eastern coasts of the continents.
- Most are surrounding islands (e.g. Japan, British Isles and Iceland).

(Or other acceptable answers)

(b) Fishing boats with a large size:

Fishermen can catch fish further from shore / hold more fish before they need to return to shore. (1)

Advanced technology for locating shoals of fish:

Fishermen use technology such as radar and sonar, which can locate shoals of fish and increase their yield. (1)

Bigger nets in use:

Fishermen can trap larger numbers of fish, including immature fish. (1)

(c) (i) There was a drop in the total weight of fish caught in the early 1990s. (1)

The fish population has declined. This shows that the fish population is harvested at a higher rate than it can recover. (1)

(ii) Any ***three*** of the following: (1 mark each) (3)

- Promoting aquaculture
- Enforcing fishing moratorium
- Setting up conservation laws
- Banning destructive fishing practices
- Restriction on net types
- Restriction on net sizes

(Or other acceptable answers)

(iii) Any ***two*** of the following: (1 mark each) (2)

- Decrease in yield leads to decrease in the income of fishermen.
- There is an increasing demand for fish for animal feed and human food.
- It is difficult to monitor / patrol in the seas and fishing grounds.

(Or other acceptable answers)

(13 marks)

##

	Skill assessed	Mark(s)
(a)	Analysing data	3
(b)	Applying concepts	3
(c)(i)	Analysing data	1
	Understanding basic concepts	1
(c)(ii)	Understanding basic concepts	3
(c)(iii)	Understanding basic concepts	2

|!|BCELC0360000023|!|

(a) ***Figure I*** shows the changes in the total human population in the world, and the estimated population in the future.

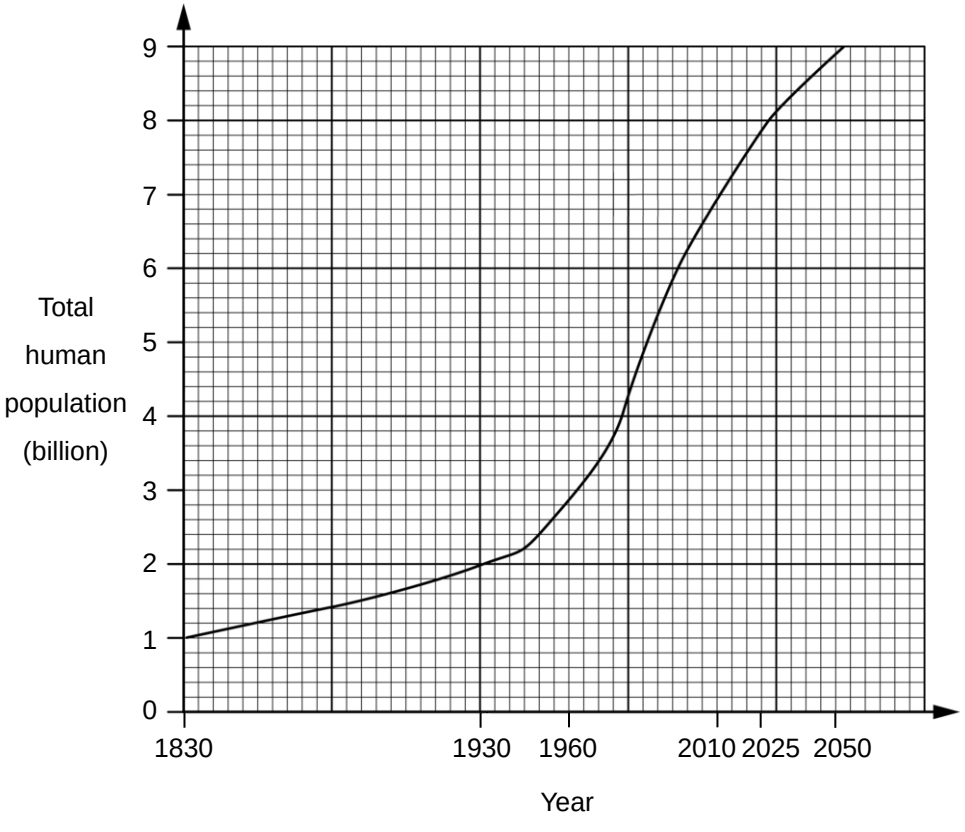


Figure I

According to ***Figure I***, describe the changes in the world human population in the past and in the future. (3 marks)

(b) World population growth increases the pressure on the Earth's natural resources. The time taken to replenish the annual consumption of animals, crops and biomass by humans is 14.3 months.

How does this information show an unsustainable use of the Earth's natural resources? (2 marks)

(c) ***Figure II*** shows the percentage changes in the area of forest in the world and different continents from 1990 to 2000.

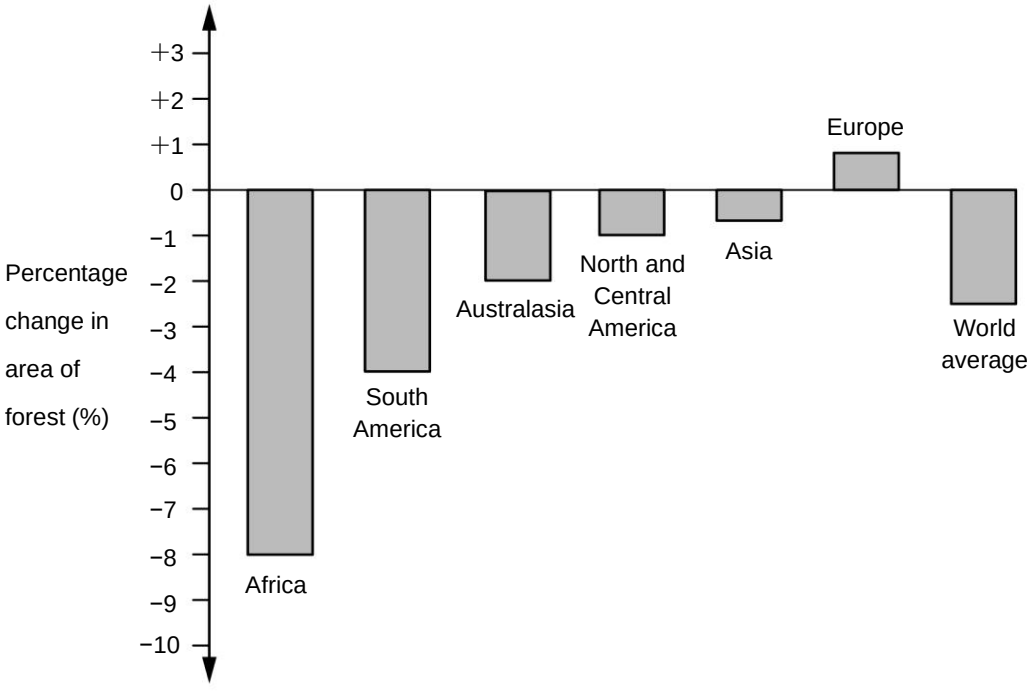


Figure II

(i) With reference to ***Figure II***, which continent shows a positive change in the area of forest? (1 mark)

(ii) Suggest ***two*** reasons to explain why more forests are lost in some continents than in others. (2 marks)

(iii) State ***two*** environmental impacts of forest loss. (2 marks)

##

(a) From 1830 to 1945, there was a steady and slow population growth. (1)
From 1945 to 2010, there was an exponential population growth. (1)
For the future, there will be a gentler increase in population. (1)

(b) Since the replenishment time of natural resources (i.e. 14.3 months) is longer than the consumption time of natural resources (i.e. 12 months), (1)
the natural resources will be depleted eventually and less resources will be available for future generations. (1)

(c) (i) Europe (1)

(ii) Any ***two*** of the following: (1 mark each) (2)

- Many forests are cleared to provide timbers in those continents with rich forest resources.
- The forests are cleared to provide land for farming in those continents with many developing countries.
- Forests are cleared due to rapid development / urbanization in those continents with many developing countries.
- Forests are greatly conserved. / There is reforestation of trees in those continents with more developed countries.

(Or other acceptable answers)

(iii) Any ***two*** of the following: (1 mark each) (2)

- Increased soil erosion
- Global warming caused by greenhouse effect
- Loss of habitats for wildlife

(Or other acceptable answers)

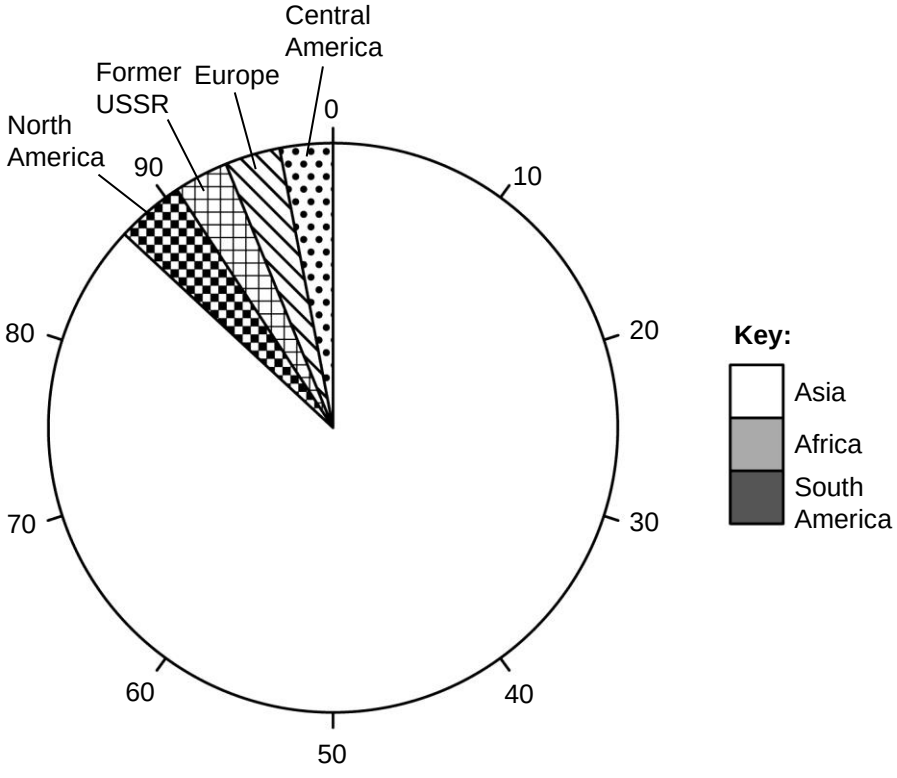
(10 marks)

##

	Skill assessed	Mark(s)
(a)	Analysing data	3
(b)	Understanding basic concepts	2
(c)(i)	Analysing data	1
(c)(ii)	Understanding basic concepts	2
(c)(iii)	Understanding basic concepts	2

|!|BCELC0360000024|!|

The following pie chart shows the distribution of global fuelwood production in 1998.



(a) Using the key above and the table below, complete the pie chart above. (2 marks)

Continent	Percentage of fuelwood production (%)
South America	11
Africa	24
Asia	52

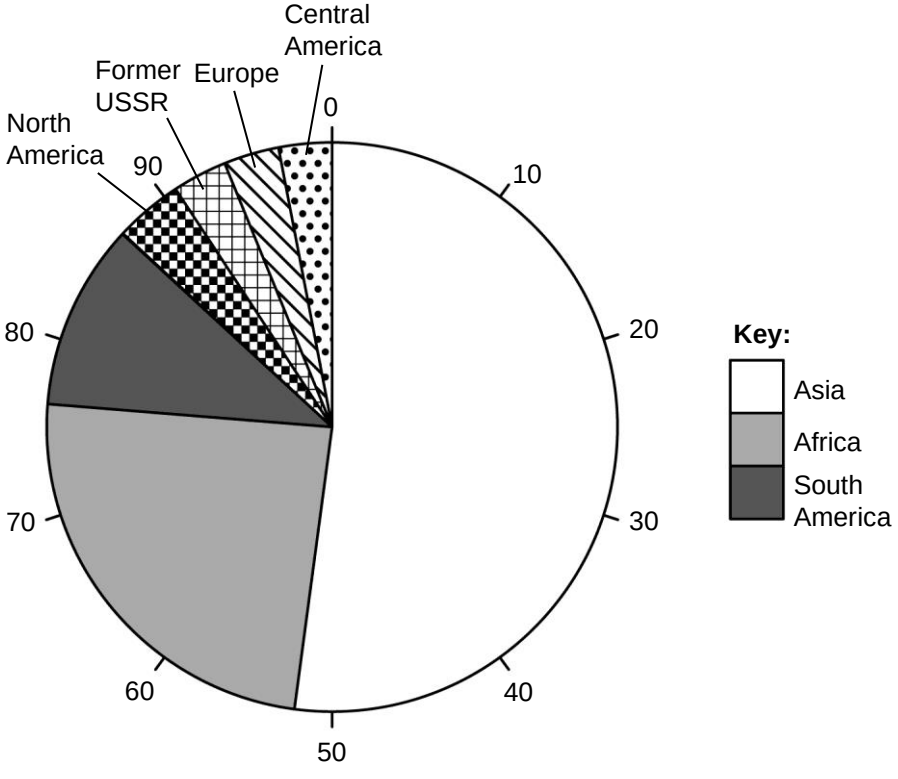
(b) What is the difference between the amount of fuelwood used in the developing world and that in the developed world? Explain why. (2 marks)

(c) State ***two*** environmental problems that will be arisen from an increase in the use of fuelwood. (2 marks)

(d) Suggest ***two*** ways that could make the use of fuelwood more sustainable.(2 marks)

##

(a)



Correct shading of the portions (1)

Correct distribution of the portions (1)

(b) The amount of fuelwood used in the developing world is much greater than that in the developed world. (1)

This is because the developed world comprises just 13% of the global fuelwood

production while the developing world contributes to 87% of the global fuelwood production. (1)

(c) Any ***two*** of the following: (1 mark each) (2)

- More cutting down of trees will contribute to global warming as fewer trees can absorb the carbon dioxide (a greenhouse gas) in the air.
- Soil will become more susceptible to soil erosion as it is not covered and protected by the trees.
- Removal of trees destroys the habitats of many plants and animals.

(Or other acceptable answers)

(d) Any ***two*** of the following: (1 mark each) (2)

- Use those dead branches as fuelwood.
- Grow trees to replace those trees that have been cut down.
- Limit the extent of deforestation for fuelwood by legislation.

(Or other acceptable answers)

(8 marks)

##

	Skill assessed	Marks
(a)	Presenting information	2
(b)	Analysing data	2
(c)	Understanding basic concepts	2
(d)	Understanding basic concepts	2

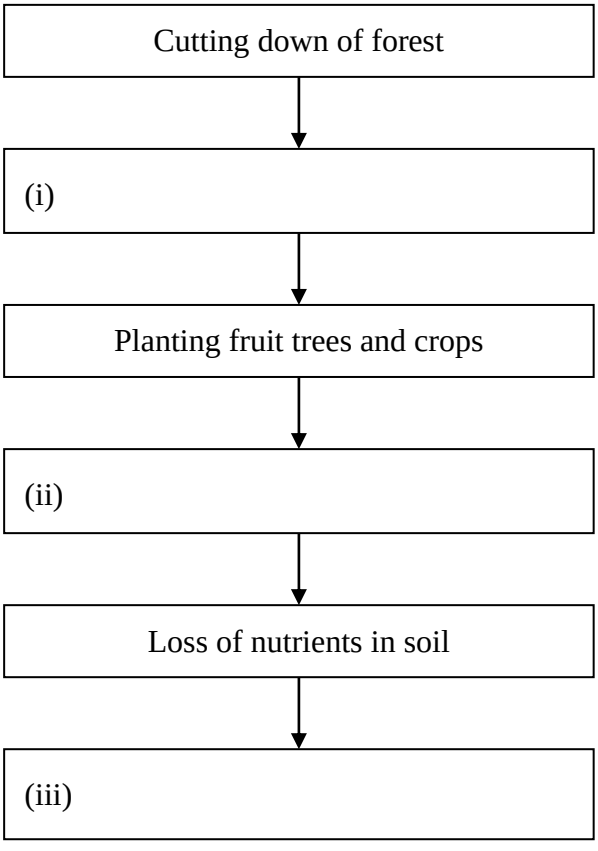
|!|BCELB0360000025|!| ★ **STSE** **Cross topic**

(a) The following paragraph shows a description of the lives of Kayapo people.

> The Kayapo people live in the Amazon rainforest. They are subsistence farmers who adopt shifting cultivation. The Kayapo people first clear the land by cutting down trees and burning the vegetation. They then plant their fruit trees and crops on the cleared land. After about four years, the heavy rains have washed away the soil nutrients. The Kayapo then move away to a new area to grow crops. The forest clearance is small enough to allow the forest to replenish in the abandoned land within a few years.

With reference to the above description of shifting cultivation, complete the flowchart below to show the sequence of events in the shifting cultivation.

(3 marks)



(b) Suggest ***two*** reasons to explain why the Kayapo people burn the vegetation before planting their crops. (2 marks)

(c) (i) Suggest why the shifting cultivation described above may be regarded as sustainable agriculture. (2 marks)

(ii) Explain why human population growth may make shifting cultivation an unsustainable agriculture. (3 marks)

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(d) In order to grow crops, trees have to be cleared from the land.

(i) State ***one*** possible effect of the clearing of trees on the wildlife in Amazon rainforest. (1 mark)

_____

(ii) Suggest ***two*** ways to reduce this effect on the wildlife. (2 marks)

_____

_____

##

(a) (i) Burning of the vegetation (1)

(ii) Heavy rains (1)

(iii)Move away to a new area to grow crops (1)

(b) It can make space for fruit plants or crop plants to grow. (1)

The ashes formed from burning can enrich the soil / act as fertilizer. (1)

(c) (i) The cleared land can be abandoned to form forest within a period of time. (1)

The forest formed can then be cleared to grow crops. (1)

(ii) Population growth means a greater demand for food / land. (1)

People thus have to clear larger areas of forests (1)

before the forests are fully recovered to the original state. (1)

(d) (i) Habitat loss / food shortage / reduced biodiversity (1)

(ii) Any ***two*** of the following: (1 mark each) (2)

- Replanting / reforestation
- Setting up of conservation areas / nature reserves

(Or other acceptable answers)

(13 marks)

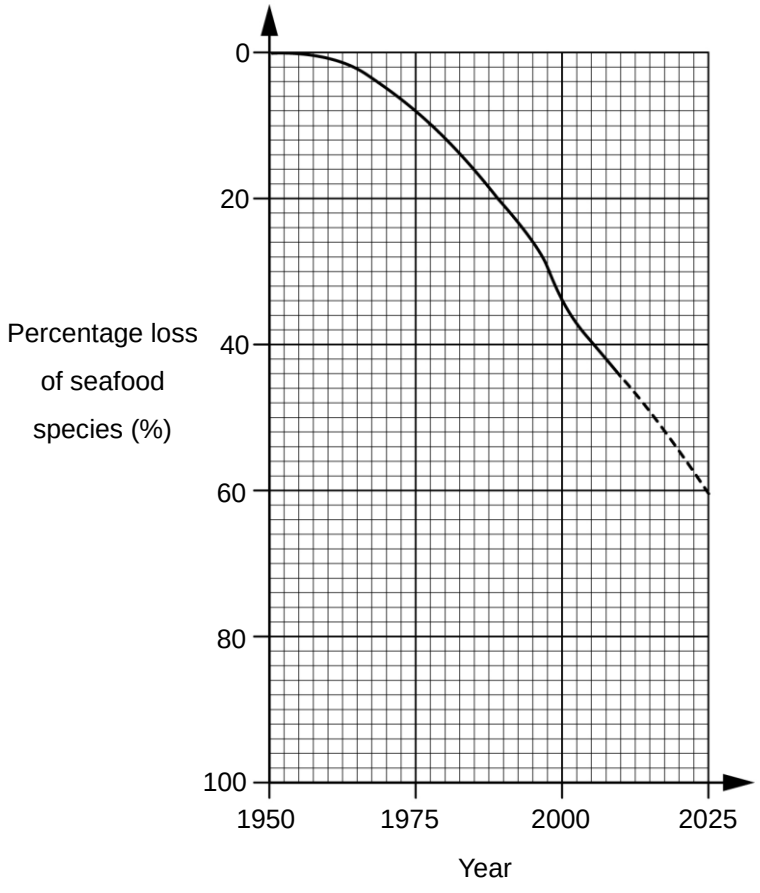
##

	Skill assessed	Mark(s)
(a)(i)	Understanding basic concepts	1
(a)(ii)	Understanding basic concepts	1
(a)(iii)	Understanding basic concepts	1
(b)	Understanding basic concepts	2
(c)(i)	Applying concepts	2
(c)(ii)	Applying concepts	3
(d)(i)	Understanding basic concepts	1
(d)(ii)	Understanding basic concepts	2

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|!|BCELC0360000026|!| **Cross topic**

The following graph shows the global percentage loss of seafood species and the expected trend in the future.



(a) (i) What is the percentage loss of seafood species from 1950 to 2010? (1 mark)

_____

(ii) State ***two*** reasons to explain for the loss of seafood species since 1950. (2 marks)

_____

_____

_____

(b) State ***two*** impacts of the loss of seafood species on the ecosystem. (2 marks)

_____

_____

(c) Suggest ***two*** measures to achieve sustainable fishing. (2 marks)

_____

_____

##

(a) (i) 44% (1)

(ii) Any ***two*** of the following: (1 mark each) (2)

- Overfishing causes the fish population difficult to replenish.
- Marine pollution (e.g. oil spillage) kills the fish.
- Human activities damage the marine habitat and hence lead to reduction in fish population.

(Or other acceptable answers)

(b) Any ***two*** of the following: (1 mark each) (2)

- The predators of fish may die due to the shortage of food.
- The prey of fish may increase in number.
- Biodiversity decreases.

(Or other acceptable answers)

(c) Any ***two*** of the following: (1 mark each) (2)

- Using nets with larger mesh / using nets with smaller size
- Enforcement of the fishing moratorium
- Restricted zones for fishing
- Restriction on the number of fishing boats
- Promoting aquaculture
- Setting up of fish taxes

(Or other acceptable answers)

_____

(7 marks)

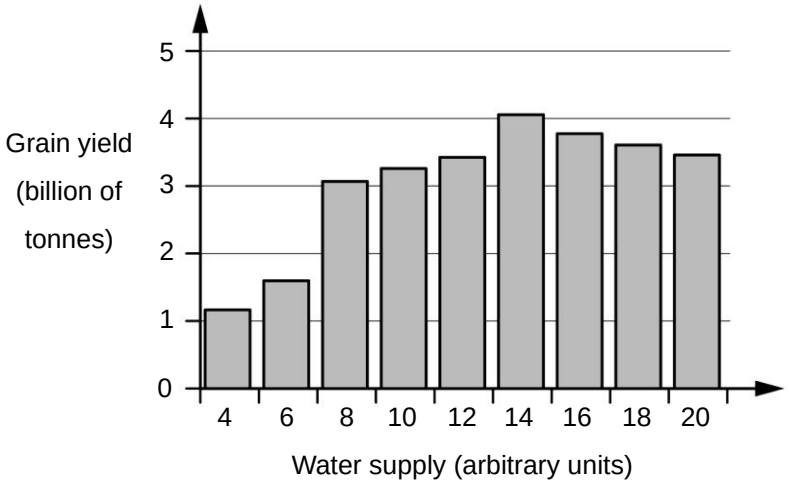
##

	Skill assessed	Mark(s)
(a)(i)	Analysing data	1
(a)(ii)	Understanding basic concepts	2
(b)	Understanding basic concepts	2
(c)	Understanding basic concepts	2

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|!|BCELC0360000027|!|

The following graph shows the effect of water supply on the yield of grain growing in a desert area.



(a) (i) How much water should be supplied to the crop for the best yield? (1 mark)

(ii) Explain why adding too much water to the soil reduces the grain yield. (3 marks)

(b) (i) State ***two*** factors that contribute to the desertification of land. (2 marks)

(ii) Explain why the ***two*** factors in (b)(i) contribute to desertification of land. (2 marks)

##

(a) (i) 14 arbitrary units (1)

(ii) The pore spaces between soil particles are filled up with water. (1)

The supply of oxygen to the root of the plant is reduced. (1)

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Less mineral salts are absorbed from the soil by active transport which requires energy from respiration. (1)

(b) (i) Any ***two*** of the following: (1 mark each) (2)

- Overgrazing
- Frequent ploughing
- Deforestation

(Or other acceptable answers)

(ii) Any ***two*** of the corresponding reasons: (1 mark each) (2)

- Overgrazing: When the grass is removed, the soil is exposed and can erode easily.
- Frequent ploughing: The topsoil is loosened, allowing the soil to be easily blown away by wind or washed away by water.
- Deforestation: the soil is not covered by the tree canopies and is easily blown away by wind.

(Or other acceptable answers)

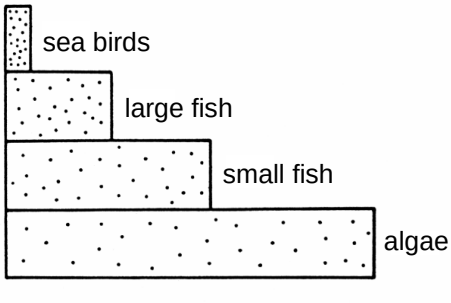
(8 marks)

##

	Skill assessed	Mark(s)
(a)(i)	Analysing data	1
(a)(ii)	Understanding basic concepts	3
(b)(i)	Understanding basic concepts	2
(b)(ii)	Understanding basic concepts	2

|!|BCELB0360000028|!| ★ **Cross topic**

The following diagram shows a pyramid of biomass and the concentration of mercury in the organisms along a food chain. The area of each rectangle represents the total biomass of organisms at that trophic level in the food chain. The number of dots in each rectangle indicates the relative amount of mercury.



(a) Explain why the biomass of large fish is greater than that of sea birds. (3 marks)

(b) Explain why the concentration of mercury is the highest in the bodies of sea birds. (4 marks)

(c) Mercury is a pollutant. Suggest a source of this pollutant. (1 mark)

(d) If the sea birds eat the small fish instead of the large fish (with the amount of fish consumed remains constant), how would their biomass and the concentration of mercury in their bodies be different? Why? (3 marks)

##

(a) Sea birds occupy a higher trophic level. (1)
Energy is lost owing to respiration / excretion / egestion (1)
at each trophic level. (1)

(b) Mercury cannot be degraded by an organism and accumulates in the body. (1)
The predator usually feeds on a large number of prey and retains the mercury they contain. (1)
At each stage in a food chain, the predator will accumulate higher concentrations of the mercury than its prey. (1)
The higher up the food chain, the more concentrated the mercury become. (1)
So, the concentration of mercury is the highest in the bodies of sea birds, the top consumers.

(c) From chemical factories (Or other acceptable answers) (1)

(d) The biomass of sea birds would be greater and (1)

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the concentration of mercury in their bodies would be lower (1)
because of a shorter food chain. (1)

(11 marks)

##

	Skill assessed	Mark(s)
(a)	Understanding basic concepts	3
(b)	Applying concepts	4
(c)	Understanding basic concepts	1
(d)	Applying concepts	3

|!|BCELC0360000029|!|
Read the passage below and answer the questions that follow.

> Algae are the producers in the world's oceans. When they are produced in vast quantities, they can provide food for primary consumers. Red tide, which appears pink or red in colour, is actually caused by an algal bloom. In fish farms, the presence of red tide seriously affects the growth of fish at night.

(a) Describe how algae can provide energy for primary consumers. (3 marks)

(b) Give an example of primary consumers in an ocean. (1 mark)

(c) State and explain the effect of red tide on the growth of fish at night. (3 marks)

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(d) Explain why algal blooms may occur in a freshwater stream near an agricultural land. (2 marks)

##

(a) Algae undergo photosynthesis to form glucose. (1)
They change solar energy to chemical energy, (1)
which is transferred to primary consumers when the algae are ingested by them. (1)

(b) Fish / zooplankton (Or other acceptable answers) (1)

(c) At night, algae carry out respiration but do not carry out photosynthesis. (1)
There is a net uptake of oxygen by the algae and the amount of dissolved oxygen in water decreases. (1)
The fish cannot get enough oxygen and may die. (1)

(d) Chemical fertilizers leached from an agricultural land increase the nutrient content in the freshwater stream. (1)
Excess amount of nutrients favours the growth of algae and lead to algal blooms. (1)

(9 marks)

##

	Skill assessed	Mark(s)
(a)	Understanding basic concepts	3
(b)	Understanding basic concepts	1
(c)	Understanding basic concepts	3
(d)	Applying concepts	2

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|!|BCELB0360000030|!| ★ **STSE**

Read the passage below and answer the questions that follow.

> Chlorofluorocarbons (CFCs) are organic compounds used in manufacturing industry. After being discharged into the atmosphere, CFCs can break down the ozone layer. They can remain active in the atmosphere for many years. As a result, human has a higher risk of developing skin cancer. An international agreement has been signed by some countries to stop the damage to the ozone layer.

(a) State ***one*** application of CFCs in industry. (1 mark)

_____

(b) State and explain the significance of the ozone layer to human health. (2 marks)

_____

_____

(c) Scientists discover that there is a large 'ozone hole' over Antarctica.

(i) Explain why an 'ozone hole' is formed. (1 mark)

_____

(ii) Explain why there are more cases of skin cancer in countries in the southern hemisphere than the northern hemisphere. (2 marks)

_____

_____

(iii) Suggest ***two*** ways to slow down the enlargement of the 'ozone hole'. (2 marks)

_____

_____

##

(a) Any ***one*** of the following: (1)

- As a coolant in refrigerators / air conditioners
- For making foam plastics / aerosol propellants

(Or other acceptable answers)

(b) The ozone layer can absorb much of the harmful UV light from the Sun. (1)

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As a result, less mutation occurs in human skin and so the risk of skin cancer is reduced. (1)

(c) (i) CFCs break down the ozone layer. (1)

(ii) UV light can pass through the 'ozone hole' and reach the Earth's surface. (1)

UV light is a harmful radiation which is known to increase the incidence of skin cancer in humans. (1)

(iii) Any ***two*** of the following: (1 mark each) (2)

- To reduce the emission of CFCs from factories
- To find alternative compounds to replace CFCs
- To reduce the use of foam plastics / aerosol products

(Or other acceptable answers)

(8 marks)

##

	Skill assessed	Mark(s)
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	2
(c)(i)	Understanding basic concepts	1
(c)(ii)	Applying concepts	2
(c)(iii)	Understanding basic concepts	2

|!|BCELC0360000031|!| **STSE** **Cross topic**

Pollution is a problem caused by urbanisation and industrialisation. In Hong Kong, the emission of air pollutants worsens the air quality. The table below shows the source and the amount of some air pollutants emitted from human activities in Hong Kong in 2012.

Air pollutants	Pollution source	Amount emitted (tonnes)
Carbon monoxide	Road transport	68,300
Sulphur dioxide	Public electricity generation, navigation	32,700
Nitrogen oxides	Road transport, navigation	115,000
RSPs	Road transport, navigation	6,130

(a) Which air pollutant is produced with the greatest amount from human activities? State ***one*** environmental impact of this pollutant. (2 marks)

(b) Describe how carbon monoxide affects the gas exchange in humans. (3 marks)

(c) Which air pollutant can cause acid rain? Explain how this pollutant leads to acid rain. (2 marks)

(d) According to the pollution source, suggest ***three*** ways to improve the air quality. (3 marks)

##

(a) Nitrogen oxides (1)
It can lead to photochemical smog / acid rain. (1)
(b) Carbon monoxide binds to haemoglobin to form carboxyhaemoglobin. (1)
The combination is irreversible. (1)
This reduces the oxygen-carrying capacity of the blood. (1)
(c) Sulphur dioxide / nitrogen oxides (1)
Sulphur dioxide/ nitric oxides react(s) with water vapour and oxygen in the atmosphere to form sulphuric acid / nitric acid. (1)
The acid makes the rainwater acidic.
(d) Any ***three*** of the following: (1 mark each) (3)
- Replace diesel vehicles with vehicles that use liquefied petroleum gas
- Switch off idling engines
- Install catalytic converters in old vehicles
. - Increase the cleaner energy source (e.g. natural gas) in electricity generation
(Or other acceptable answers)

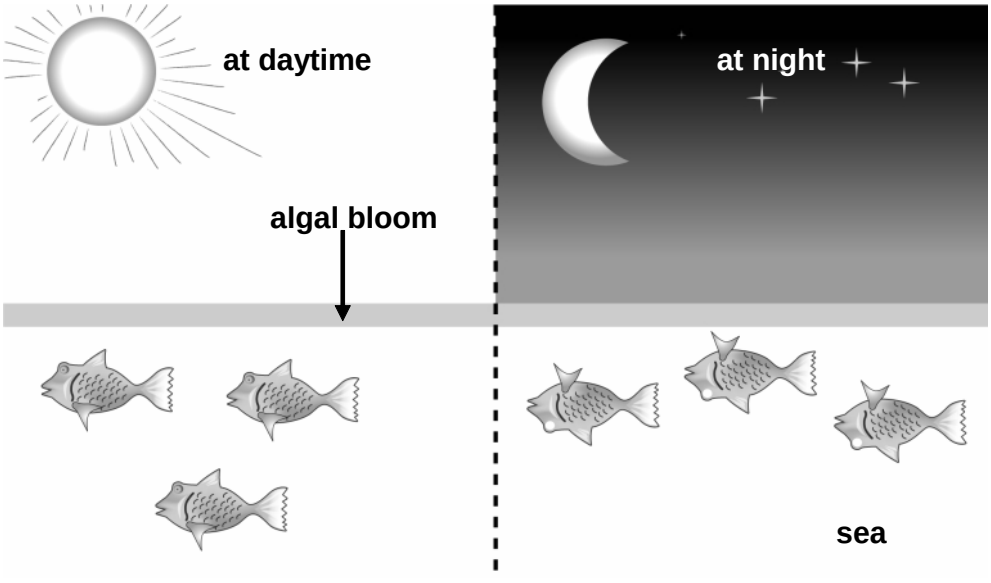
(10 marks)

##

	Skill assessed	Mark(s)
(a)	Analysing data	1
	Understanding basic concepts	1
(b)	Understanding basic concepts	3
(c)	Understanding basic concepts	2
(d)	Understanding basic concepts	3

|!|BCELC0360000032|!| **Cross topic**

The diagram below shows the effects of algal bloom on fish at daytime and at night.

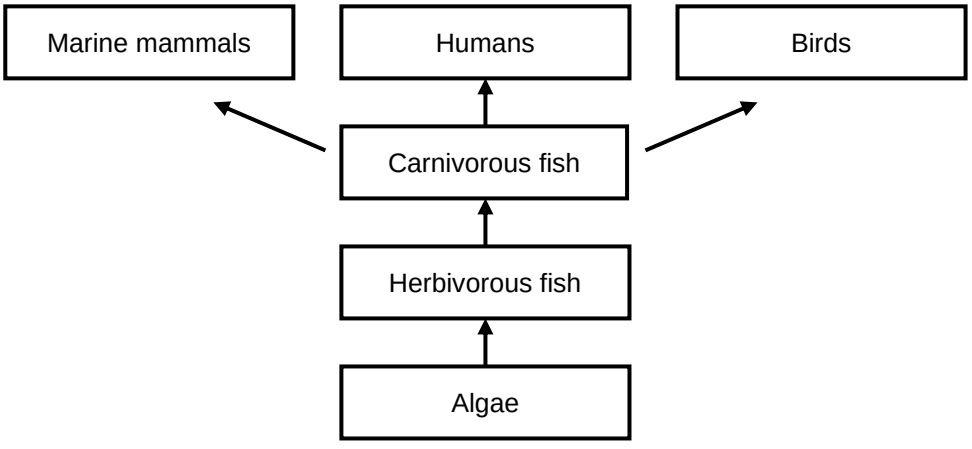


(a) Explain the cause of an algal bloom. (2 marks)

(b) When algal bloom occurs, many fish die at night. Explain why. (3 marks)

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(c) The following figure shows a food web.



(i) What is the relationship between humans, marine mammals and birds in the food web? (1 mark)

(ii) Some algae species produce toxins. Explain why the risk of food poisoning is much higher when humans eat carnivorous fish when compared with herbivorous fish. (4 marks)

##

(a) If there is an excess amount of nutrients (e.g. nitrates) in water, (1)
algae will grow rapidly and increase in population near the water surface, resulting in algal bloom. (1)

(b) At night, the algae carry out respiration and do not carry out photosynthesis. (1)
Most of the oxygen in water is depleted. (1)
The fish under water may die of suffocation / from lack of oxygen. (1)

(c) (i) (interspecific) competition (1)

(ii) The toxins produced by algae are accumulated throughout the food web. (1)
They can neither be broken down nor excreted by the body. (1)
As carnivorous fish feed on a large number of herbivorous fish, the amount of toxins is much higher in the bodies of carnivorous fish. (1)
The toxin level in the carnivorous fish is so high that they will cause food poisoning when humans consume the fish. (1)

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(10 marks)

##

	Skill assessed	Mark(s)
(a)	Understanding basic concepts	2
(b)	Understanding basic concepts	3
(c)(i)	Analysing data	1
(c)(ii)	Understanding basic concepts	4

|!|BCELB0360000033|!| ★ **Cross topic**

Visibility is generally considered poor when it is below 8 km. The table below shows the percentage of days with visibility impairment due to air pollution from 1991 to 2000 in Hong Kong.

Year	**1991**	**1992**	**1993**	**1994**	**1995**	**1996**	**1997**	**1998**	**1999**	**2000**
Percentage of days with visibility below 8 km	4.4	4.9	5.5	5.7	5.5	8.8	10.1	8.2	8.9	6.3

(Note: The relative humidity must be less than or equal to 75%)

(a) Using the data above, plot a graph below to show the changes in the percentage of days with visibility below 8 km from year 1991 to year 2000. (4 marks)



(b) Explain why the data should not include the days with the relative humidity above 75%. (2 marks)

(c) Suggest possible reasons to explain the changes in the percentage of days with poor visibility during the period of 1995 to 1997. (3 marks)

(d) Poor visibility is caused by the presence of photochemical smog.

(i) State ***two*** air pollutants that lead to the formation of photochemical smog. (2 marks)

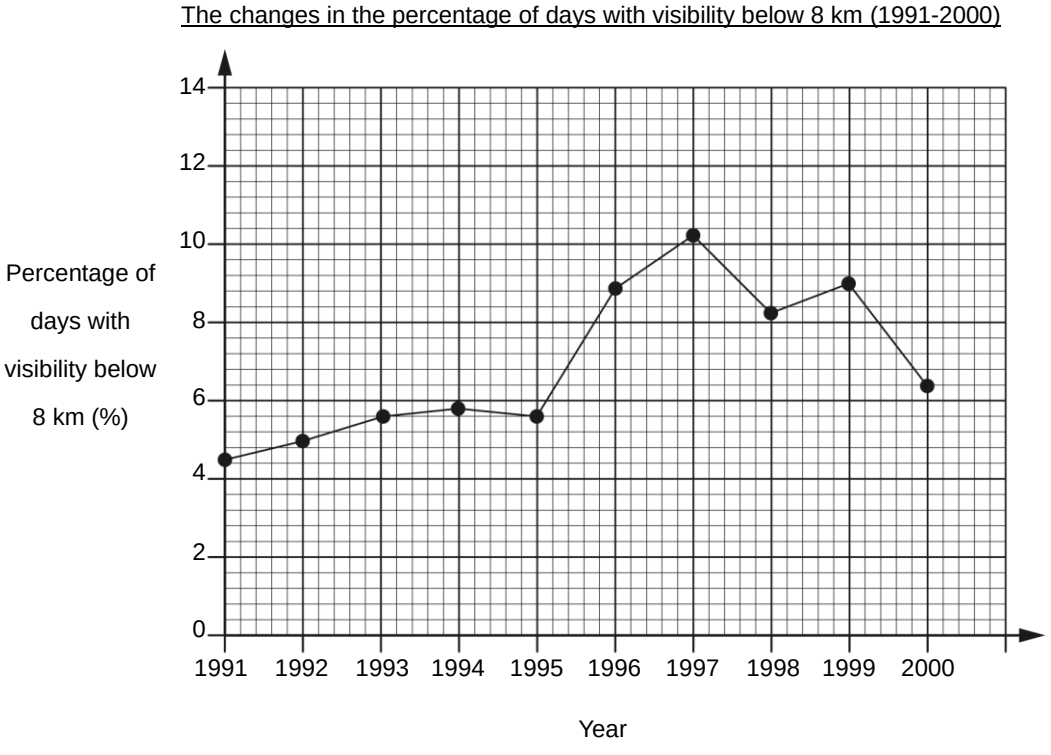
(ii) What are the effects on people who live for a long period of time in an environment with photochemical smog? State any ***two***. (2 marks)

(e) Suggest and explain ***one*** possible way in which you can do in daily life to help improve visibility in Hong Kong. (2 marks)

##

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(a)



Correct plotting and joining of points (2)
Correct choice and labelling of axes (1)
Correct title (1)

(b) When the relative humidity is higher than 75%, raining and fogging are more likely to occur. (1)
Raining and fogging can decrease the visibility, regardless the level of air pollutants. (1)

(c) From 1995 to 1997, the amount of nitrogen oxides in the air increases (1)
because of the increased amount of exhaust fumes emitted from vehicles / industries / power plants. (1)
Nitrogen oxides lead to the formation of photochemical smog which decreases the visibility. (1)

(d) (i) Nitrogen oxides (1)
Volatile organic compounds (e.g. hydrocarbons) (1)
(ii) Any ***two*** of the following: (1 mark each) (2)
- It can irritate the eyes and the breathing system.
- It can affect lung function, causing shortness of breath and coughing.
- It can trigger asthma.
(Or other acceptable answers)

(e) Use public transport whenever possible. (1)

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This can help reduce production of pollutants (e.g. nitrogen oxides) from motor vehicles. (1)

(Or other acceptable answers)

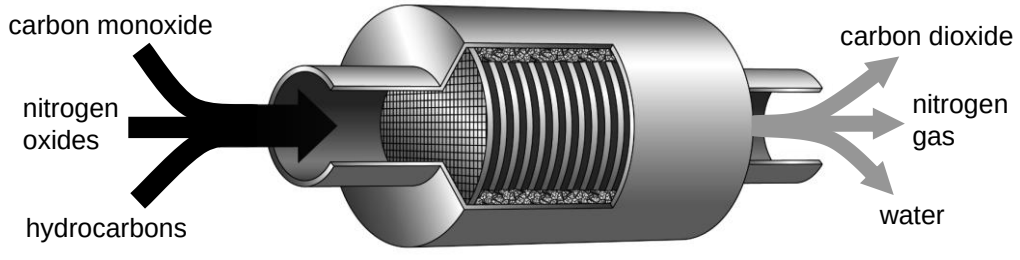
(15 marks)

##

	Skill assessed	Mark(s)
(a)	Presenting information	4
(b)	Practical skills	2
(c)	Analysing data	1
	Understanding basic concepts	2
(d)(i)	Understanding basic concepts	2
(d)(ii)	Understanding basic concepts	2
(e)	Understanding basic concepts	2

|!|BCELB0360000034|!| ★

The diagram below shows a catalytic converter which is a device installed in the exhaust pipe of a car in order to reduce the amount of air pollutants emitted.



(a) How do the nitrogen oxides formed? (1 mark)

(b) According to the diagram, why can the catalytic converter reduce the amount of pollutants emitted into the air? (1 mark)

(c) What are the adverse effects of carbon monoxide and nitrogen oxides on health? State ***one*** adverse effect for each. (2 marks)

(d) Increase in the number of vehicles will cause a global environmental issue.

(i) Name this environmental issue. (1 mark)

(ii) How does an increase in the number of vehicles lead to this environmental issue? (3 marks)

##

(a) Nitrogen oxides are formed when fuels are burned at high temperatures. (1)

(b) The catalytic converter can convert the air pollutants (e.g. nitrogen oxides, hydrocarbons) in car exhausts into harmless substances (e.g. water, nitrogen, carbon dioxide). (1)

(c) Adverse effect of carbon monoxide: reduce the oxygen-carrying capacity of blood (1)

Any ***one*** of the adverse effects of nitrogen oxides: (1)
- Irritate the breathing system
- Increase the risks of respiratory infections / chronic lung diseases
- Irritate the eyes

(d) (i) Global warming (1)

(ii) More greenhouse gas (e.g. carbon dioxide, nitrous oxides, etc.) is emitted from the increased number of vehicles. (1)

The greenhouse gas can absorb some heat emitted from the Earth's surface. (1)

This makes the atmosphere become warmer and increases the average global temperature. (1)

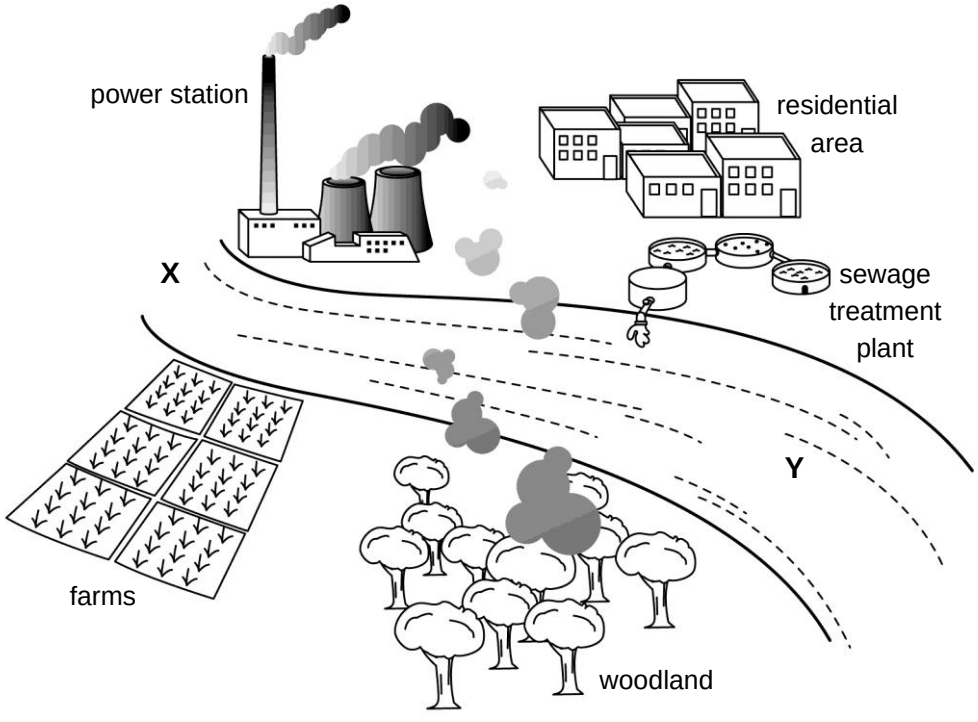
(8 marks)

##

	Skill assessed	Mark(s)
(a)	Understanding basic concepts	1
(b)	Applying concepts	1
(c)	Understanding basic concepts	2
(d)(i)	Understanding basic concepts	1
(d)(ii)	Applying concepts	3

|!|BCELB0360000035|!| ★

The diagram below shows the map of a city.



(a) In which site, X or Y, is more suitable for fishing? Explain your answer.(3 marks)

(b) The exhaust fumes from the power station lead to the formation of acid rain. Acid rain affects the growth of trees in the woodland.

(i) Name ***one*** substance in the exhaust fumes that forms acid rain. (1 mark)

(ii) State ***two*** adverse effects caused by acid rain on the growth of trees.(2 marks)

(c) If the sewage from the residential area is not treated by the sewage treatment plant and drained into the river directly, what will be the change in the dissolved oxygen level of the river? Explain your answer. (4 marks)

(d) Farmers make use of DDT as an insecticide. State and explain the effect of DDT on the fish-eating birds. (6 marks)

##

(a) Site X (1)

This is because site Y is heavily polluted by the discharge of sewage from the power station, farms and the residential area. (1)

There will be fewer fish in site Y. / The fish may be poisonous in site Y. (1)

(b) (i) Sulphur dioxide / nitrogen oxides (1)

(ii) Any ***two*** of the following: (1 mark each) (2)

- Acid rain destroys the leaf cuticle and the leaves become more susceptible to bacterial and fungal infections.
- Acid rain lowers the soil fertility and inhibits the growth of trees.

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- Acid rain enhances the release of heavy metal ions from the soil and high level of these ions can damage the roots of trees.

(Or other acceptable answers)

(c) The residential sewage contains a high level of organic matter. (1)
Bacteria decompose the organic matter and their number increases rapidly. (1)
The bacteria will use up oxygen in the water during decomposition. (1)
The dissolved oxygen level of the river thus decreases. (1)

(d) DDT is washed off from the farmland into the river. (1)
DDT is taken up by the water plants. (1)
Since DDT cannot be broken down and cannot be excreted, (1)
DDT remains in the bodies of water plants and is transferred to the fish body when the plants are ingested. (1)
Since fish-eating birds consume a large number of fish during its lifetime, DDT is accumulated to a higher concentration in the birds. (1)
The level of DDT is high enough to kill the birds. (1)

(16 marks)

##

	Skill assessed	Mark(s)
(a)	Applying concepts	1
	Understanding basic concepts	2
(b)(i)	Understanding basic concepts	1
(b)(ii)	Understanding basic concepts	2
(c)	Applying concepts	4
(d)	Applying concepts	6

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|!|BCELC0360000036|!|

Acid rain causes serious environmental problems in some parts of the world. As acid rain enters a lake, the lake becomes acidic.

Figure 1 shows how acid rain is formed and how it affects the pH of a lake.

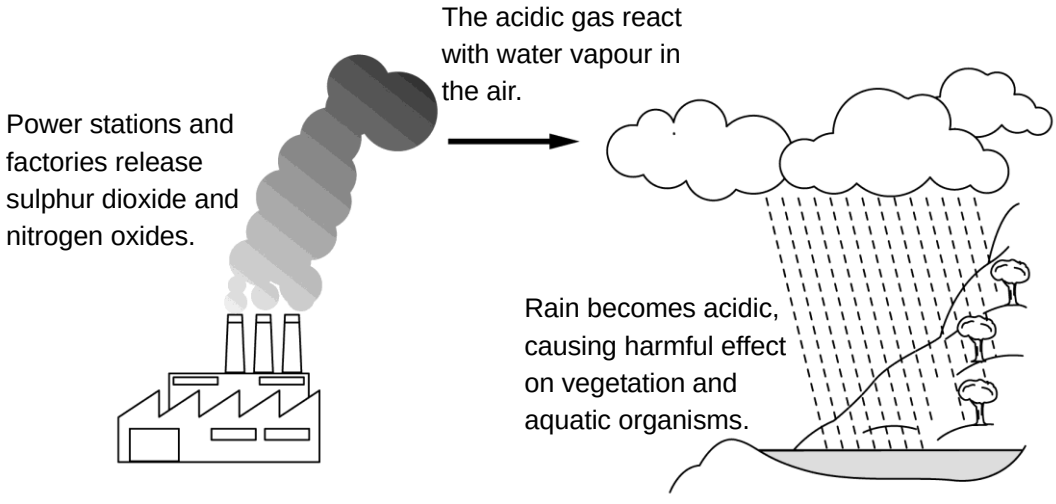


Figure 1

(a) (i) Besides the source of acidic gas as shown in ***Figure 1***, state another source of acidic gas. (1 mark)

_____

(ii) State ***two*** harmful effects of acid rain on the forest. (2 marks)

_____

_____

_____

(b) With reference to the source of acidic gas, suggest ***two*** ways to reduce acid rain. (2 marks)

_____

_____

_____

Figure 2 shows the pH range in which some animals living in lakes can tolerate. The shaded boxes represent the pH value at which the animals can survive.

Animals	**pH 6.5**	**pH 6.0**	**pH 5.5**	**pH 5.0**	**pH 4.5**	**pH 4.0**	**pH 3.5**
Blackfly larvae	■	■	■	■	■	■	■
Mayfly larvae	■	■	■				
Clams	■	■					
Trouts	■	■	■	■			
Basses	■	■	■				
Frogs	■	■	■	■	■	■	
Salamanders	■	■	■	■			

Figure 2

(c) Use the information provided in ***Figure 2*** to answer the following questions.

(i) Which animals can survive in the lake with pH as low as 4.0? (1 mark)

_____

(ii) Which animals are the most sensitive to a drop in pH? (1 mark)

_____

(iii) Explain why some aquatic animals cannot tolerate at a pH lower than 4.0. (2 marks)

_____

_____

_____

(d) State ***two*** harmful effects of sulphur dioxide and nitrogen oxides on human health. (2 marks)

_____

_____

_____

##

(a) (i) Any ***one*** of the following: (1)

- Vehicle exhausts
- Combustion of fossil fuels in factories

- Volcanic eruptions

(Or other acceptable answers)

(ii) Any ***two*** of the following: (1 mark each) (2)

- Acid rain destroys the leaf cuticle and the leaves become more susceptible to bacterial and fungal infections.
- Acid rain lowers the soil fertility and inhibits the growth of trees.
- Acid rain enhances the release of heavy metal ions from the soil and high level of these ions can damage the roots of trees.

(Or other acceptable answers)

(b) Any ***two*** of the following: (1 mark each) (2)

- Use renewable energy sources, e.g. wind power / tidal power / solar power
- Use fuels with low sulphur content
- Reduce the use of coal
- Use more public transport
- Use electric cars

(Or other acceptable answers)

(c) (i) Frogs and blackfly larvae (1)

(ii) Clams (1)

(iii) Enzymes do not function well / denature at low pH. (1)

Acid damages the shells / scales / skin (external tissues) of animals. (1)

(d) Any ***two*** of the following: (1 mark each) (2)

- They irritate the eyes.
- They irritate the breathing system.
- They increase the risk of having chronic lung diseases.

(Or other acceptable answers)

(11 marks)

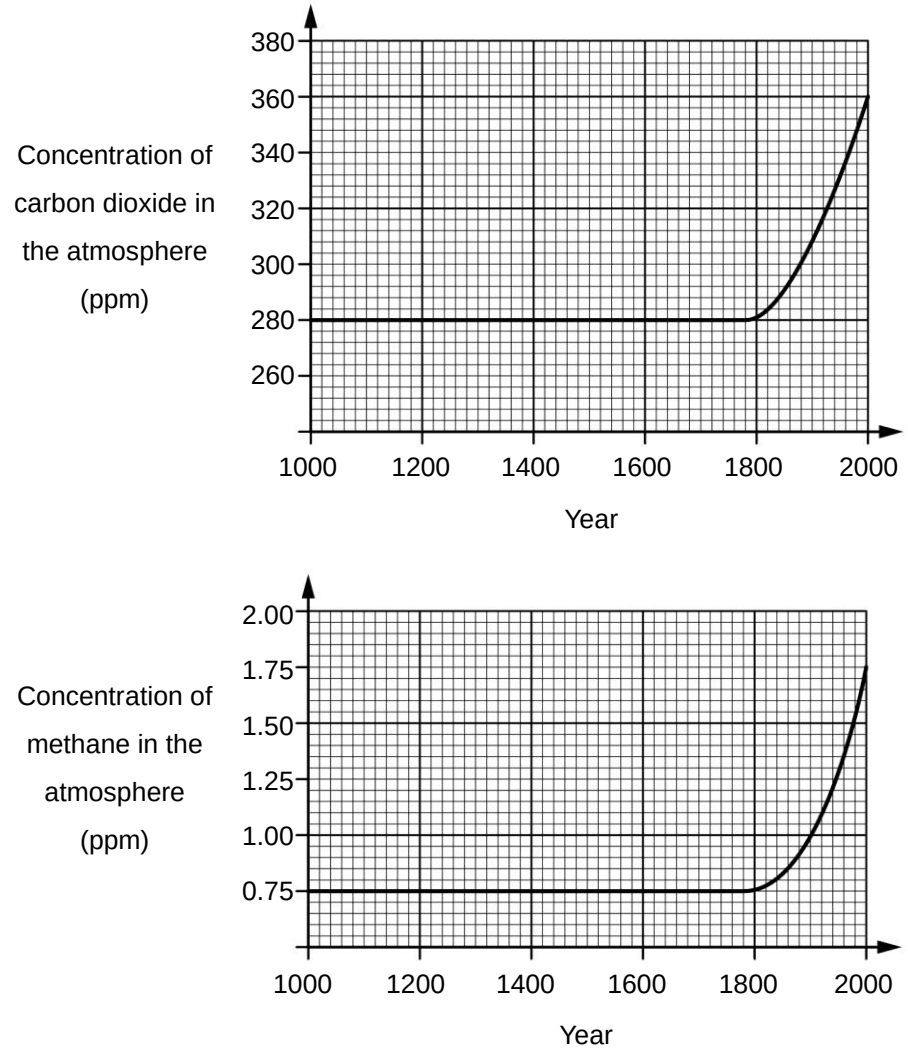
##

	Skill assessed	Mark(s)
(a)(i)	Understanding basic concepts	1
(a)(ii)	Understanding basic concepts	2
(b)	Understanding basic concepts	2
(c)(i)	Applying concepts	1
(c)(ii)	Applying concepts	1
(c)(iii)	Understanding basic concepts	2
(d)	Understanding basic concepts	2

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|!|BCELB0360000037|!| ★

Carbon dioxide and methane are two of the main greenhouse gases. Human activities increase the concentration of greenhouse gases and this enhances the greenhouse effect. The graphs below show the changes in the concentrations of carbon dioxide and methane in the atmosphere over the past 1000 years.



(a) According to the above graphs, describe the changes in the concentrations of carbon dioxide and methane over the past 1000 years. (2 marks)

(b) Explain the changes you described in (a). (4 marks)

(c) Explain how carbon dioxide and methane contribute to global warming.(3 marks)

(d) State and explain ***two*** harmful effects of global warming. (4 marks)

##

(a) The concentrations of both gases in the atmosphere were relatively constant before 1800. (1)

There was a sharp increase in the concentrations of both gases from 1800 to 2000. (1)

(b) Any ***two*** of the following causes for the increase in carbon dioxide concentration: (1 mark each) (2)

- Increased combustion of fossil fuels released more carbon dioxide into the atmosphere.
- Extensive deforestation decreased the number of trees, so that less carbon dioxide was absorbed by trees.

(Or other acceptable answers)

Any ***two*** of the following causes for the increase in methane concentration: (1 mark each) (2)

- More methane was released from cattle that were increasing in number.
- Methane was released from rice fields that were increasing in area.
- Increased waste disposal caused more methane released from anaerobic

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breakdown in landfill sites.

(Or other acceptable answers)

(c) The greenhouse gases trap part of the heat that is emitted from the Earth's surface after the absorption of solar radiation by the Earth's surface. (1)

As these greenhouse gases increase in concentrations in the atmosphere, more heat is trapped in the atmosphere. (1)

As a result, the average temperature of the atmosphere increases. (1)

(d) Any ***two*** sets of the following: (2 marks each) (4)

1. Increased melting of icebergs / thermal expansion of sea water leads to a rise in sea level.

   This causes flooding in low-lying coastal areas / extinction of species due to loss of habitats.

2. The frequency of extreme weather events increases.

   This will lead to loss of life and property / reduction of agricultural productivity.

3. A warmer climate will favour the spread of pathogens.

   More people may suffer from infectious diseases.

(Or other acceptable answers)

(13 marks)

##

	Skill assessed	Marks
(a)	Analysing data	2
(b)	Understanding basic concepts	4
(c)	Understanding basic concepts	3
(d)	Applying concepts	4

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|!|BCELB0360000038|!| ★

Most aquatic species can survive in water of pH between 6 and 7 only. A reduction of the pH to less than 6 can be harmful to them. ***Figure 1*** shows the pH ranges in which four common river invertebrates can survive. These invertebrates feed at the lower trophic levels of the food chain in the river.

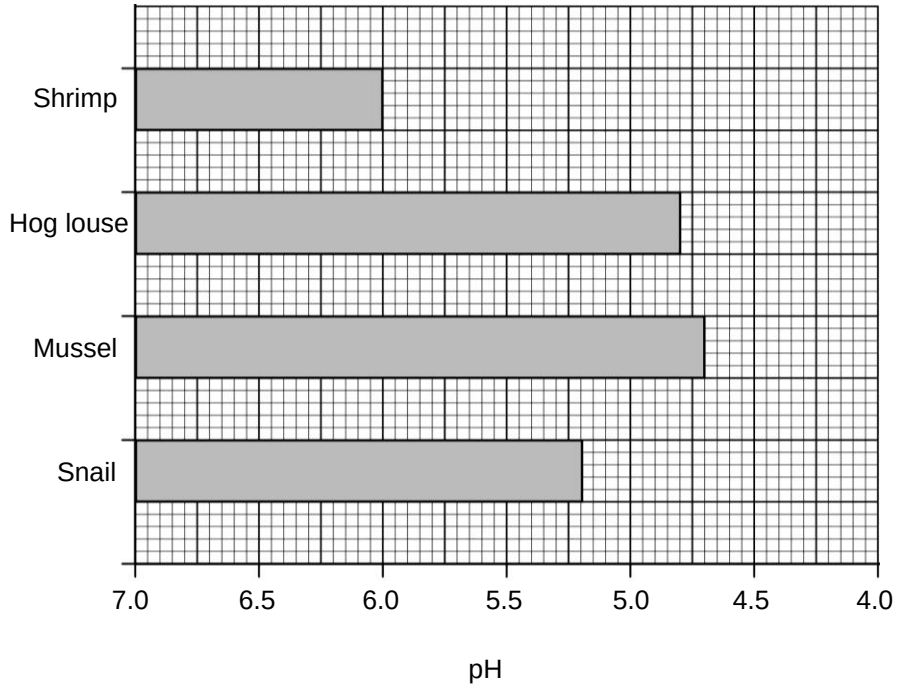


Figure 1

(a) Some air pollutants contribute to the acidification of rivers. Name ***one*** of the air pollutants and explain how it may cause the acidification of rivers. (4 marks)

(b) According to ***Figure 1***, suggest which invertebrates may increase in number when the river has become acidified. Explain your answer. (3 marks)

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(c) ***Figure 2*** shows the annual catches of salmons in rivers of pH above 5 (non-acidified rivers) and in rivers of pH below 5 (acidified rivers) between 1935 and 1985. The catches in each year are relative to the catches of 1935. (Catches in 1935 = 100%)

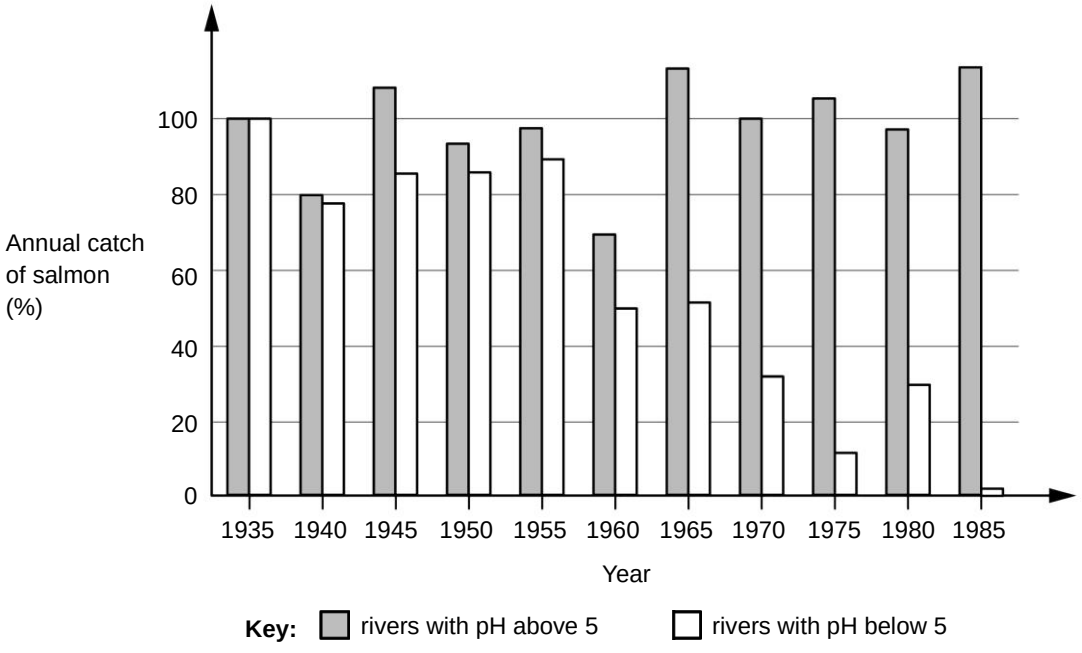


Figure 2

According to ***Figure 2***, state the changes in the annual catches of salmons in non-acidified rivers and acidified rivers between the years of 1935 and 1985.

(2 marks)

(d) Suggest why an increase in river acidity may lead to a fall in the population of fish.

(2 marks)

##

(a) Sulphur dioxide / nitrogen oxides (1)

This acidic gas reacts with water vapour and oxygen in the atmosphere to form an acid (sulphuric acid / nitric acid). (1)

This acid makes rainwater more acidic. (1)

When acid rain falls into river, it lowers the pH of the river. (1)

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(b) Mussel and hog louse (1)

When the river has become acidified, there is less competition for food / space. (1)

Many of their predators cannot survive in acidified river. (1)

As a result, they increase in number in acidified rivers.

(c) Catch of salmon increased slightly in non-acidified rivers. (1)

Catch of salmon decreased drastically in acidified rivers. (1)

(d) Any ***two*** of the following: (1 mark each) (2)

- The development of fish eggs is affected by acidic water; so fewer fish can hatch.
- Acid can damage the scales / skin of fish.
- Enzymes denature and lose their functions in the presence of acids.

(Or other acceptable answers)

(11 marks)

##

	Skill assessed	Mark(s)
(a)	Understanding basic concepts	4
(b)	Analysing data	1
	Understanding basic concepts	2
(c)	Analysing data	2
(d)	Applying concepts	2

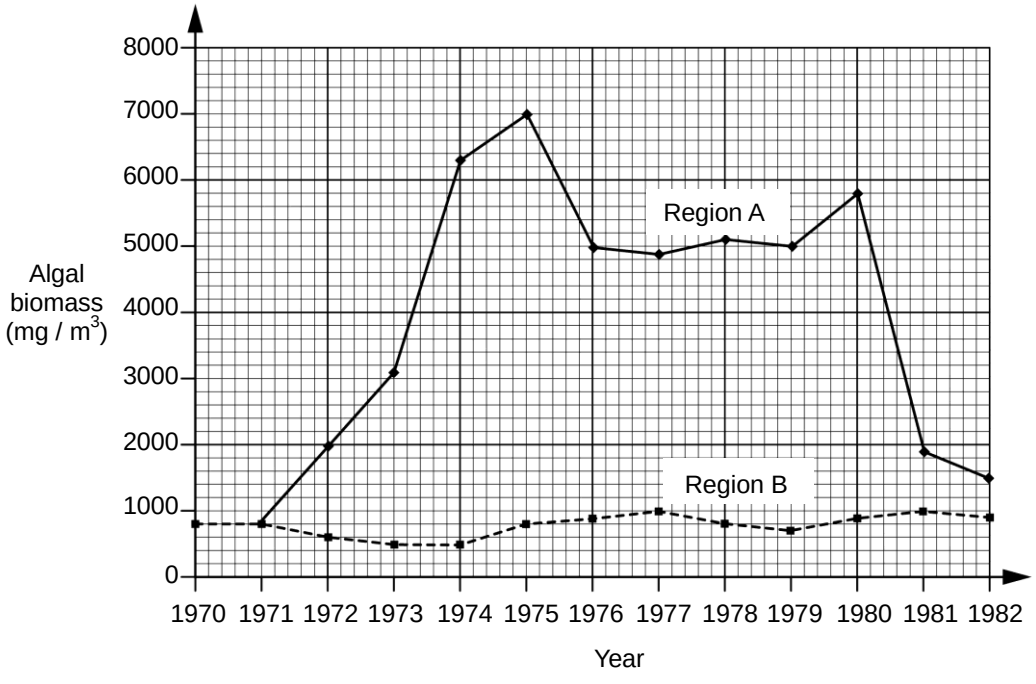
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|!|BCELB0360000039|!| ★ **SI**

Scientists carried out an investigation in 1970 to study the growth of algae in a lake. They first divided a lake into two equal regions (A and B) using a plastic curtain. The two regions had the same biomass of algae at the start of the investigation.

From 1970 to 1975, they added extra nitrates into the water of Region A every year. After 1975, they stopped adding extra nitrates in that region. Region B acted as the control and no extra nitrates were added throughout the investigation.

The biomass of the algae in the water in both regions was measured each year and the results are shown in the following graph.



(a) (i) According to the above graph, calculate the percentage increase in the algal biomass in Region A between 1971 and 1975. Show your working. (2 marks)

(ii) State and explain the changes in the algal biomass in Region A between 1971 and 1976. (5 marks)

(b) (i) What is the term used to describe the rapid increase in algal biomass in the Region A? (1 mark)

(ii) State and explain ***two*** effects on other organisms living in the lake when extra nitrates were added into the lake. (4 marks)

##

(a) (i) $(7000 - 800) \div 800 \times 100\%$ (1)

$= 775\%$ (1)

(ii) The algal biomass increased from 1971 to 1975. (1)

This was because nitrate was used for the synthesis of amino acid and this helped increase the growth rate. (1)

The algal biomass decreased from 1975 to 1976. (1)

This was because stopping supplying nitrates to water limited the growth of algae. (1)

Algae died and were consumed by other animals. Algal biomass thus decreased. (1)

(b) (i) Algal bloom (1)

(ii) Any ***two*** sets of the following: (2 marks each) (4)

1. Increase in algal biomass increased the amount of food for primary consumers.
   Organisms of higher trophic levels also increased in number.
2. Algal bloom blocked the sunlight for other plants.
   Other plants could not carry out photosynthesis and thus died.
3. Algae released toxins.
   The toxins killed animals / other organisms.
4. After large amounts of algae have died, they were decomposed by bacteria which consumed the oxygen in the water.
   The depletion of oxygen in the water led to suffocation of aquatic organisms.

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(Or other acceptable answers)

(12 marks)

##

	Skill assessed	Mark(s)
(a)(i)	Mathematical skills	2
(a)(ii)	Analysing data	2
	Applying concepts	3
(b)(i)	Understanding basic concepts	1
(b)(ii)	Applying concepts	4

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